Investigating Stakeholder Influence in the Co-design of VR Game Experiences for Intangible Heritage

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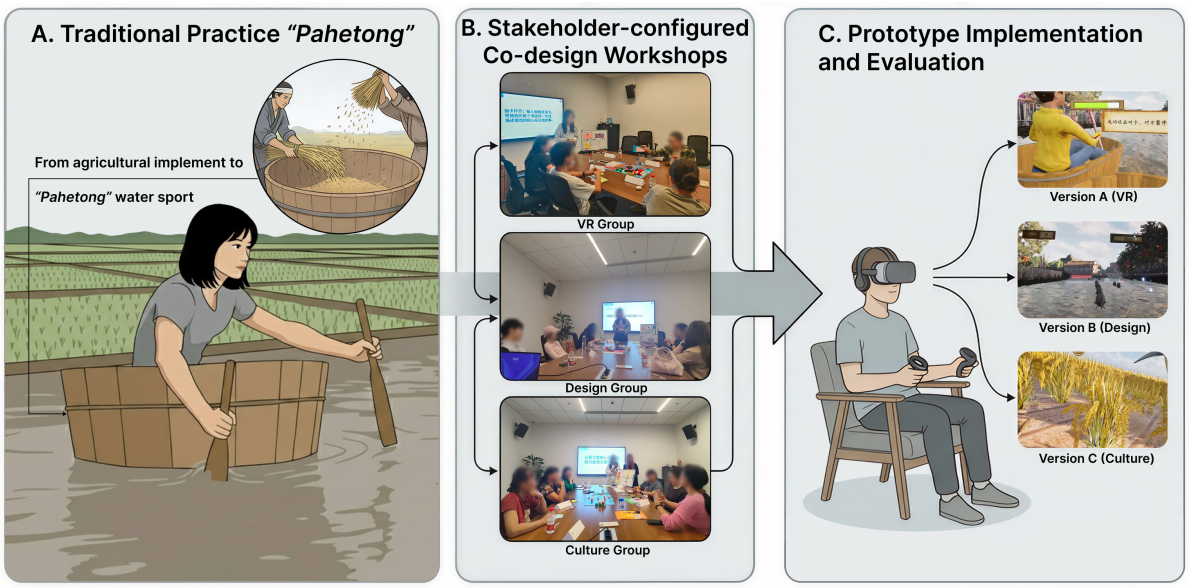

Fig. 1. A. Pahetong: from a rice-harvesting grain bucket to a local water-based sport. B. Stakeholder-configured co-design workshops with three groups (VR, Design, Culture). C. Three corresponding VR prototypes implemented from workshop concepts and evaluated in a within-subjects user study.

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Co-design is increasingly adopted in virtual reality (VR) applications for intangible cultural heritage (ICH), yet the influence of stakeholder composition on both process and outcomes remains insufficiently examined. This paper investigates how different configurations of cultural familiarity, design experience, and VR literacy shape the co-design of a VR game based on the traditional ICH practice of *"Pahetong"*. Across three structured workshops, we analysed participation roles, knowledge contributions, and decision-making dynamics, and developed three corresponding VR prototypes, which were evaluated in a within-subjects user study (N = 24). Results show that while structured authorship can initially distribute participation, role differentiation during convergence and prototyping strongly influences which ideas become operational mechanics. Moreover, cultural authenticity alone does not guarantee interpretability; legibility depends on how tacit practices are translated into feasible and learnable interactions. We contribute empirical insights into stakeholder influence in heritage VR co-design and propose design considerations for transforming culturally grounded knowledge into playable, interpretable digital experiences.

CCS Concepts: • **Applied computing** → **Computer games**; • **Human-centered computing** → **Virtual reality**; **Participatory design**.

Additional Key Words and Phrases: Co-design, intangible cultural heritage, virtual reality, gaming, digital heritage, user experience evaluation

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1 Introduction

Virtual reality (VR) has become an important medium for digital cultural heritage, enabling immersive, interactive, and embodied experiences that go beyond conventional screen-based presentations [18, 29]. In heritage education, VR is used to convey the spatial, historical, and affective dimensions of places and practices, and is often promoted as a way to reach broader and younger audiences through gamified and exploratory forms of learning [25, 83]. Furthermore, VR-based embodied experiences can be extended to older adults by emphasizing perceived usefulness, ease of use, enjoyment, and culturally familiar content [76]. Within this broader landscape, intangible cultural heritage (ICH) such as rituals, traditional arts, and embodied crafts has become a central focus. Recent systems on Wuhu iron painting, traditional Chinese flower arrangement, and VR learning environments for multiple ICH practices show that carefully designed VR experiences can support both skill development and reflection on cultural meaning [72, 77, 84]. Yet there remains a limited understanding of how different interaction strategies in VR can simultaneously mediate between the past and the future of cultural heritage while enhancing user experience and educational impact [47].

In response, HCI and digital heritage researchers increasingly argue that meaningful VR heritage experiences require careful attention to the design process [44] and systematic engagement with cultural stakeholders as co-designers [3]. Participatory and co-design traditions highlight that including those affected by technology can foster mutual learning, redistribute design authority, and align design outcomes with situated values [46]. In the cultural heritage domain, co-design projects with museums, communities, and designers have explored how interpretations of authenticity, accessibility, and innovation are negotiated when creating interactive installations and VR experiences [1, 3]. Calvi's framework for stakeholder involvement in immersive cultural heritage technologies further formalizes the roles of different actors across the production chain and argues for early and continuous co-creation to support community ownership and clearer problem framing [10]. More recent work on user-centric and community-based VR heritage design extends this view by proposing concrete co-design methodologies for cultural tourism and education [30, 48].

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Building on this progress, although many studies on digital cultural heritage design consider stakeholder involvement [17, 45, 51, 74], few examine how different stakeholder configurations shape the dynamics and outcomes of co-design processes. Comparative work often varies tools and methods rather than systematically varying stakeholder composition. Outside heritage, co-design research suggests that diversity can both widen the design space and introduce communication challenges and power asymmetries that shape whose perspectives are materialised in the final design [46]. However, these insights have rarely been operationalised in controlled comparative studies in VR ICH design.

This paper responds to these gaps by examining stakeholder influence in the co-design of VR game experiences for intangible heritage. We focus on the traditional ritual of *Pahetong* and adopt a controlled comparative design. *Pahetong* is a locally distinctive ICH practice in which the wooden grain bucket (*hetong*), originally used to carry harvested rice grains, was repurposed during the off-season into water-based recreation and eventually competitive racing (see Figure 1.A). We chose *Pahetong* because it is relatively little-known yet culturally specific, and its core embodied interaction is straightforward to translate into VR gameplay. Before a series of three co-design workshops, potential participants completed a structured questionnaire that captured their degree of cultural familiarity with *Pahetong*, prior creative design experience, and VR literacy. Using these profiles, we assembled workshop groups that balanced internal diversity while varying stakeholder configurations across sessions in a systematic way. Design concepts emerging from each workshop were developed into VR game prototypes that embody different interpretations of the ritual. These prototypes were then evaluated in a within-subjects user study that combined quantitative and qualitative measures of user experience and perceived cultural relevance.

These aims are reflected in the following two research questions:

RQ1: *How does stakeholder configuration in the co-design process of a VR intangible cultural heritage game influence the dynamics of the design process*?

RQ2: *How do different stakeholder configurations affect the perceived cultural relevance and user experience of the resulting VR game prototypes*?

Building on this design and evaluation, our results show that stakeholder configuration shapes both how co-design unfolds and how cultural knowledge becomes operationalised in VR gameplay. While structured authorship initially supported distributed participation, role differentiation during convergence influenced which ideas were translated into concrete mechanics. We further find that cultural authenticity alone does not ensure interpretability. Prototypes grounded in lived experience still required additional framing to become legible to newcomers, whereas design-mediated versions more effectively balanced playability and cultural cues. Quantitatively, the three prototypes were perceived differently in terms of spatial presence, perceived competence, perceived value/usefulness, and perceived completeness, revealing trade-offs between clarity, competitiveness, and cultural specificity. By combining comparative workshops with within-subjects evaluation, this paper contributes empirical evidence on stakeholder influence in VR ICH co-design and offers actionable guidance for translating culturally grounded knowledge into playable, interpretable digital heritage experiences.

2 Background

2.1 Digital Technologies for Intangible Cultural Heritage

Unlike tangible heritage characterized by physical artifacts, ICH refers to the living culture of human communities that are continuously reconstructed in response to evolving environments [36, 52, 68], while transmitted across generations and represented in folk activities, traditional craftsmanship, artistic performances, and social practices [68, 69]. As the approach of revitalizing and disseminating ICH, digital technologies have been extensively deployed. These efforts primarily involve 3D model reconstruction [63, 66, 78] and digital media documentation

(e.g., video, audio, images, and text) [14, 22, 27, 42, 81]. For instance, Tao and Huang [66] proposed a three-dimensional visualization of the tie-dye handicraft ICH and visualized its process. Fan et al. [22] developed CICHMKG, a large-scale multimodal knowledge graph that integrates textual descriptions and images of ICH with interactive visualization interfaces.

As technology advances, VR offers a new possibility for more immersive and experiential transmission of ICH than traditional 2D screen-based interaction [29]. Existing VR-based digitization approaches often construct embodied interactive experiences by simulating the authentic scenarios and processes of cultural practices [29, 41, 61, 72, 79, 84]. Wang et al. [72] proposed FloraJing, a VR Chinese flower arrangement system to reconstruct the material and cultural environment. Zhu et al. [84] developed a VR system for Wuhu Iron Painting that enables users to experience the crafting process. Although prior studies demonstrate the efficacy of VR-based digitization in facilitating transmission of practice-oriented ICH, they emphasize high-fidelity immersion and the authenticity of practice [72, 79, 84]. Consequently, how VR-based digitization can be designed to promote active cultural engagement and sustained participation warrants further investigation.

2.2 VR Games for Cultural Engagement

Gamification refers to integrating game-related design components into non-gaming environments [20], which has proven to promote user engagement, motivation, and learning outcomes in the educational environment [4, 19, 56]. These advantages are well aligned with the goals of ICH transmission and preservation. Therefore, gamification has been increasingly merged into ICH experiences [12, 18, 24, 35, 37, 40, 54, 72, 82, 84]. ICH encompasses diverse domains, including oral traditions and expressions, performing arts, social practices, rituals and festive events, knowledge and practices concerning nature and the universe, and traditional craftsmanship [69]. Given this diversity and our focus on VR games, we organize prior work around two design directions most relevant to interactive heritage gameplay: embodied practice and performance, and narrative exploration and cultural knowledge dissemination.

One major design direction in VR-based ICH games focuses on embodied practice and performance, emphasizing hands-on craft-making and traditional performance activities. It engages in VR games through bodily interactions and participatory performances, which facilitate immersive learning and the acquisition of cultural skills [35, 41, 43, 72, 84]. For instance, Liu et al. [41] combined avatars, participatory performance, and game-based knowledge acquisition to develop a VR framework for the traditional oral performance art "Hua'er" folk songs. Moreover, Zhao et al. [43] developed a system based on body sensing equipment that allows users to imitate the Yicheng Flower Drum with full-body movements.

Furthermore, another major design direction emphasizes narrative exploration and cultural knowledge dissemination. These VR ICH games leverage interactive storytelling, role-playing, and immersive exploration to convey symbolic meanings, historical knowledge, and social contexts, thereby enhancing cultural understanding and cross-cultural empathy [18, 24, 54, 82]. For example, Paolis et al. [18] developed an immersive VR application to reconstruct the inaccessible Castle of Corsano and pop-ups or panels to introduce its history and cultural background. Petrelli et al. [54] compared VR and tangible interactions to engage visitors with Roman culture, highlighting how immersive environmental storytelling and role-playing enhance cultural comprehension.

Despite the promising applications of VR-based gamification for ICH, most studies have primarily focused on proposing a VR culture game and evaluating the final user experience of it, such as immersion, usability, or cultural perception [18, 41, 43, 54, 82]. Relatively few studies have investigated how different game design strategies influence both user experience and cultural understanding, which limits the ability to create meaningful digitalized culture experiences.

2.3 Co-designing with Diverse Stakeholders

Co-design is an essential approach to prevent cultural experiences from becoming superficial and misunderstood, supporting deep embodiment of cultural knowledge [34, 38, 73]. It allows diverse participants to contribute their knowledge to collaboratively develop solutions as equal partners instead of a single expert or designer's decision [2, 58] and is predicated on Participatory Design (PD) [65]. For instance, Liaqat et al. [39] hold co-design workshops for helping immigrant families preserve cultural depth across generations and fostering comprehension. Nevertheless, the outcomes of co-design are highly dependent on the participants and need to balance diverse requirements. Prior studies on gamified XR interaction further suggest that different user groups may require different forms of engagement, safety support, and collaboration [13, 75].

Therefore, many studies proposed involving diverse stakeholders throughout the entire co-design process and paying attention to the requirements of scarce groups [21, 23, 30, 46, 48, 73]. Mosha et al. [48] emphasized the inclusion of elderly people in the co-design process and divided stakeholders into knowledge, interest, and power groups. However, the study did not compare how different stakeholder configurations influenced collaboration outcomes. Similarly, Dolcetti [21] validated co-design in archaeology via two workshops, but did not systematically examine differences in stakeholder configurations and interaction patterns. Although these studies demonstrate the value of diverse stakeholder participation in cultural co-design, there is still a lack of systematic comparison regarding how stakeholder configurations affect collaboration dynamics, user participation, and design outcomes. This gap not only makes it difficult for other researchers to replicate or adapt the collaboration frameworks, but also obscures issues such as power imbalances, knowledge asymmetries, or the marginalization of local voices during the co-design process. Accordingly, our study investigates how stakeholder configuration influences the co-design process and outcomes of VR-based ICH game design, while providing actionable guidance for researchers, designers, and cultural institutions.

3 Co-design Methods

3.1 Participants' Recruitment

Table 1. Participant demographics and pre-workshop screening scores by group.

Group	*n*	Age bands (n)					Gender (F:M)	Bachelor+ (%)	Res. ≥1y (%)	Screening means (0–1)		
		18–24	25–34	35–44	45–54	55+				Culture	Design	VR
Culture	6	0	2	0	1	3	6:0	50%	100%	**0.637**	0.573	0.378
Design	5	3	1	1	0	0	3:2	80%	20%	0.118	**0.725**	0.680
VR	4	2	2	0	0	0	2:2	100%	50%	0.029	0.422	**0.817**

We recruited adult volunteers through local community outreach, posts on a WeChat Official Account, and Rednote. These channels have broad user bases and reliably reach publics interested in cultural heritage and interactive media. Interested individuals completed a short pre-workshop questionnaire covering three dimensions relevant to the study: familiarity with the intangible heritage context of *Pahetong*, experience in creative design, and experience with VR (see Appendix A). Items were rated on Likert scales and averaged within each dimension; scores were then normalized to the 0–1 range to enable comparison across participants.

Group assignment was based on the questionnaire data. We formed three workshops, each oriented toward one of the dimensions, so that participants could collectively contribute their strongest expertise on that facet while retaining a baseline in the other two. The three target groups were Culture-oriented, Design-oriented, and VR-oriented. The three workshops were designed as expertise-oriented stakeholder configurations rather than representative samples of broader stakeholder communities. Participants were selected to contribute different

forms of cultural, design, and VR experience, rather than to represent all possible perspectives within those categories. To minimize priming, participants were not informed of the group configuration prior to their attendance. We aimed for five participants per workshop. On the workshop day, one participant in the VR group did not attend, and a Culture-group participant arrived with an accompanying attendee who had already completed the pre-workshop questionnaire. We accepted both deviations to preserve scheduling and to avoid excluding community members who had already invested time in preparation. Final group sizes, descriptive demographics, and normalized screening means are reported in Table 1.

All procedures received institutional review board approval. Written informed consent was obtained from every participant prior to data collection, and those who completed the full set of workshop activities received a 100 RMB (approximately 15 USD) honorarium.

3.2 Co-design Process

All three groups followed the same script and facilitation materials over approximately 90 minutes, proceeding through five phases of telling, making, and enacting with a divergent–convergent rhythm [7]: (1) brief context-setting and shared references; (2) a structured game-design decomposition; (3) divergent ideation with guided convergence; (4) a rapid paper-prototype enactment of the core loop; and (5) closing reflections and consolidation (See Fig. 2). All workshops were led by a single facilitator (the first author, a designer and design researcher), ensuring consistent pacing and protocol across sessions. Workshops were conducted in Mandarin Chinese. For the Culture group, where several participants communicated primarily in Cantonese, the local language of the area, a Cantonese-speaking assistant supported facilitation, clarification, and note taking.

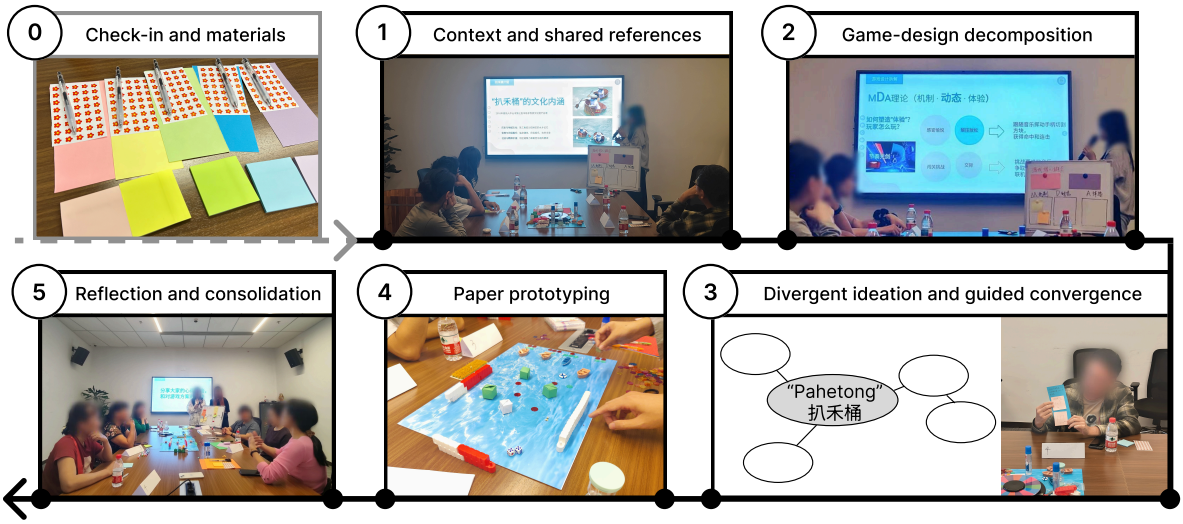

Fig. 2. After check-in and materials distribution, all three workshops followed the same 90-minute sequence of five phases: (1) Context and shared references; (2) Game-design decomposition using the MDA lens with reference VR titles; (3) Divergent ideation with guided convergence via E→P→R prompts; (4) Paper prototyping and brief enactment of the core loop; (5) Reflection and consolidation of the workshop.

3.2.1 Phase 1: Context and shared references. We began with a brief introduction and a prompt about prior awareness of *Pahetong* and VR. A concise briefing situated *Pahetong* as an intangible heritage practice and outlined its historical evolution from a farm tool to a local sport. A primer on VR interaction followed, after which we presented two canonical VR titles (Beat Saber and Moss) as accessible exemplars of distinct play paradigms. One participant briefly tried Beat Saber while others observed; additional hands-on time was offered after the session.

3.2.2 Phase 2: Game-design decomposition. We introduced the Mechanics-Dynamics-Aesthetics (MDA) lens [28] and grounded it with two reference VR games. To make the framework accessible to a mixed, nonspecialist audience, we adapted MDA into a simplified Experience–Play–Rules (E–P–R) triad, with experience, play, and rules corresponding to aesthetics, dynamics, and mechanics, respectively. Through brief interactive prompts, the facilitator first asked participants to identify the most salient player experiences in each example, then to articulate how those experiences arise from patterns of play, and finally to specify the concrete rules and feedback that sustain those patterns.

3.2.3 Phase 3: Divergent ideation and guided convergence. Participants first populated a shared pool of word cards: on individual sticky notes, each person wrote as many single words or short phrases as they could think of related to *Pahetong* (one term per note). After the notes were collected and shuffled, each participant drew two cards at random and composed a one-sentence concept that combined the two prompts. This random-prompt, high-throughput ideation activity follows established game-design brainstorming practices using word prompts and index cards [5]. The "two-word pitch" round was run twice. We then conducted silent dot voting, allocating three votes per participant, to select the top two to three one-sentence concepts as candidate cores for the game. Building on the highest-voted items, we conducted a rapid, question-prompted convergence using the E→P→R order.

- *Experience (Aesthetics):* Who is the player, what is the in-game viewpoint, and what core experience are we seeking?
- *Play (Dynamics):* What is the goal of the game, and which elements can the player interact with?
- *Rules (Mechanics):* Which inputs map to which in-game effects, and when does a round end, and on what criteria?

Each question was time-boxed to one minute per participant to maintain pace and concreteness. After each block, we clustered near-duplicate responses and, consistent with participatory design "design game" practice [6], conducted a brief vote or a short moderated discussion to agree on the working answer for that question before proceeding.

3.2.4 Phase 4: Paper prototyping. Using paper, pens, tokens, and simple desktop props, teams externalized the selected concept into a minimal, runnable model of the core loop.

3.2.5 Phase 5: Reflection and consolidation. We concluded with a reflection and consolidation segment. The facilitator summarized the game concept, and each participant shared their impressions of the workshop and provided feedback on the resulting design.

Throughout the study, we collected all written artifacts, including sticky notes and one-sentence concept statements, and we made audio and video recordings of the sessions for subsequent analysis.

3.3 Co-design Outcomes and Prototyping

3.3.1 Co-design Outcome. Three game design concepts emerged from the workshops, one from each group (See Fig. 3). Each concept specifies the player role and goal, the core loop, and the key resources or interactables that drive play.

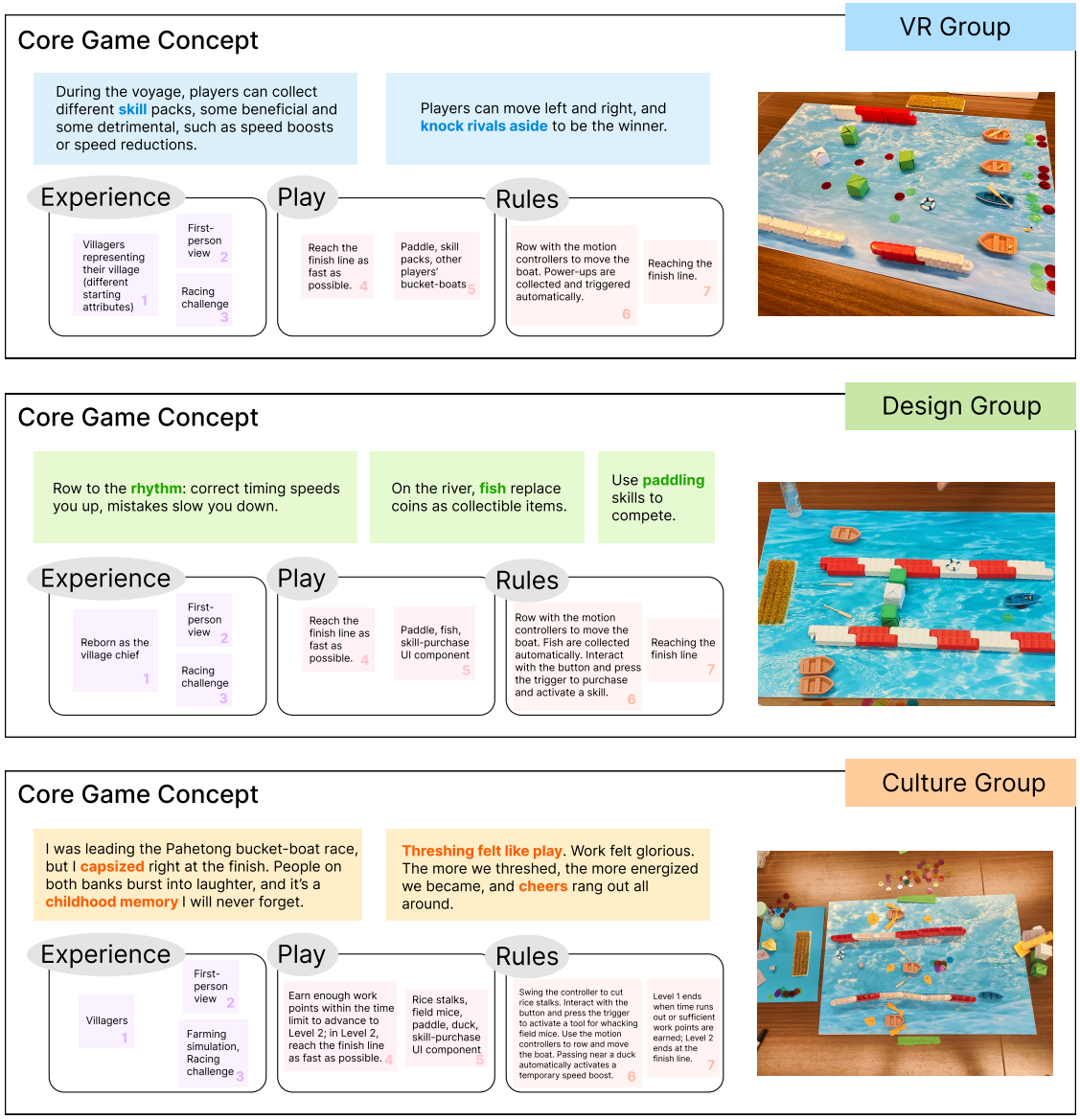

Fig. 3. Co-design outcomes by group, organized by Experience (Aesthetics), Play (Dynamics), and Rules (Mechanics). Numbers 1 to 7 correspond to: (1) player, (2) viewpoint, (3) core experience, (4) goal, (5) interactables and resources, (6) input to effect mappings, and (7) end condition. VR group: the player is a villager racing for village honor (first person); the goal is to reach the finish; interactables include paddle, health packs, random skill packs, and rival boats; input is rowing with controllers with automatic pickup of power-ups and automatic triggering of the selected skill; the round ends at the finish. Design group: the player is a reincarnated village chief (first person) who rows to calls in the local dialect; fish collected on the route are used to purchase random skills such as higher speed, more fish, or muting interference; the player can select an initial skill at the start; the round ends at the finish. Culture group: a two-level concept with Level 1 as threshing to earn work points while field mice steal rice and tools can be purchased, and Level 2 as a bucket boat race with ducks that grant short speed boosts; Level 1 ends when time expires or when the player earns sufficient work points. Level 2 ends at the finish.

VR Group. Players act as villagers racing on behalf of their village in *Pahetong*. At the start, a village is selected, and villages differ in base stamina and speed. Navigation consumes stamina, and health packs restore it. Along the route, players collect health packs and random skill packs that can accelerate or decelerate performance, and they may collide with other boats to gain an advantage. The core loop is to manage stamina, time the use of skills, and contest position against rival villages to finish first.

Design Group. Players are reincarnated as village chiefs and compete in a *Pahetong* race. Paddling follows the cadence of rowing calls in the local dialect, and short bursts of disruptive music appear at intervals. Fish spawn randomly and are collected during navigation; accumulated fish can be spent to purchase random skills. Skills include higher speed tied to different paddle forms, additional fish, and muting the disruptive music. At the outset, players choose an initial skill. The core loop couples rhythm-guided paddling, opportunistic fish collection, and timely skill purchase and activation to outpace rivals.

Culture Group. Players take the role of a villager in a two-level structure. In Level 1, the task is threshing to earn work points within a time limit while field mice steal rice; work points can be exchanged for random anti-mouse tools such as a slipper. Earning enough work points unlocks Level 2, where players race in *Pahetong*. Ducks spawn at random and can be grabbed for a temporary speed boost, with the objective of reaching the finish line as fast as possible. The core loop moves from time-bound resource gathering and protection in Level 1 to a sprint style race with intermittent boosts in Level 2.

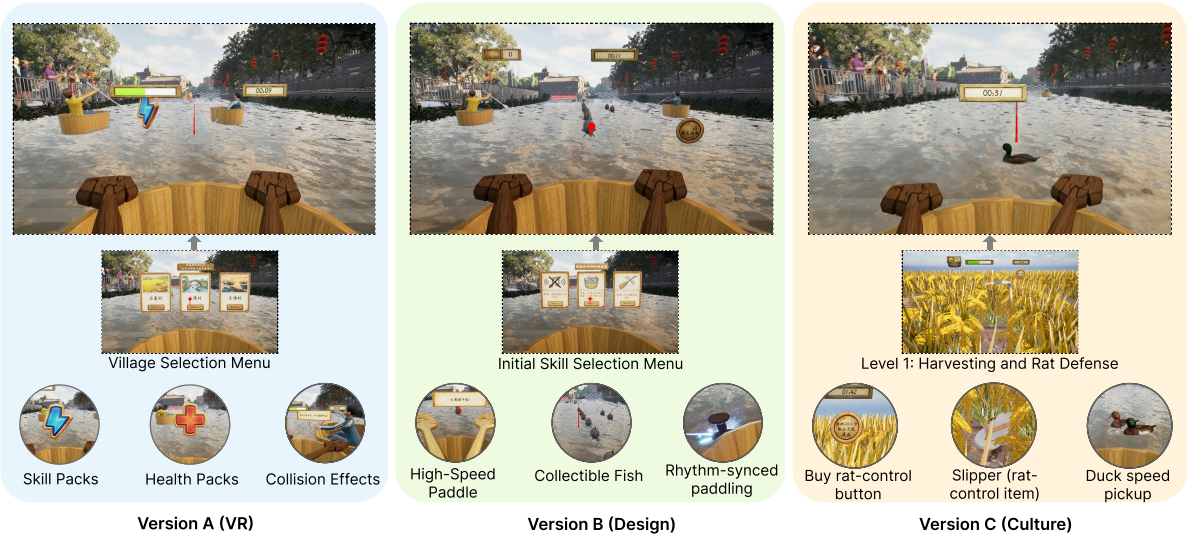

Fig. 4. Three VR game prototypes derived from the co-design workshops. Version A (VR group) shows a competitive river race with village selection, health packs, skill packs, and boat collision effects. Version B (Design group) highlights rhythm-synced paddling, collectible fish, an initial skill selection menu, and purchasable skills. Version C (Culture group) depicts a two-level structure, with Level 1: Harvesting and Rat Defense featuring a rat-control purchase button, a slipper tool for hitting rats, and duck-based speed pickups, followed by Level 2: the Pahetong river race.

3.3.2 Prototype Implementation. The prototype was developed in Unreal Engine 5 and deployed on the Meta Quest Pro. All three versions were implemented by the same development team, namely the first and second authors of this paper, who had participated throughout the workshops (see Figure 4). To ensure consistency

across conditions, the prototypes shared the same scene layout, visual assets, interaction framework, and UI style. Differences were introduced primarily in task structure and rule configuration based on the respective co-design outcomes. Audiovisual elements varied only where explicitly proposed by workshop participants, such as the rhythmic threshing chant suggested by the design group and the rice-harvesting scene emphasized by the culture group. To reduce production time and costs, some game UI assets were generated using Gemini. The shared environmental scenes across the three prototypes were created by the 3D modelling team to realistically reconstruct the local village setting, while realistic models such as rice fields and ducks were sourced from openly available online asset libraries.

4 Evaluation

4.1 Evaluation of the Co-design Process

We analysed the co-design process along three complementary dimensions, following prior work on design process analysis [8, 59]: Participation and Roles, Knowledge and Content, and Interaction and Decision-Making. For Participation and Roles, we drew on the "zones of participation" framework, which distinguishes audience, whisperer, actor, and director positions in collaborative activity [9]. In our interpretation, the audience refers to participants who are present and attentive but contribute little to decision-making; whisperers support and guide others through suggestions and brief interventions; actors actively propose ideas and move the work forward; and directors shape overall direction and make or consolidate key decisions. We identified how participants shifted between these positions across different workshop phases.

For Knowledge and Content, we conducted a content analysis of the sticky notes generated during the brainstorming phases. We coded the keywords written by participants, grouped them into higher-level categories, and compared the distribution of these categories across groups to understand what kinds of cultural, experiential, and game design knowledge were brought into play in each session.

For Interaction and Decision-Making, we relied on a joint analysis of the workshop video and audio recordings, informed by interaction analysis approaches to studying collaborative work [31]. We examined how proposals were introduced, taken up, modified, or rejected, how the facilitator and participants negotiated choices at key moments such as voting and convergence, and how concrete decisions about mechanics and rules emerged over time.

4.2 Evaluation of the Design Outcomes

4.2.1 Participants. We conducted a within-subjects evaluation study in which each participant played all three game versions (VR, Design, and Culture). We recruited 24 participants via university online forums and the Rednote platform, targeting people interested in VR-based intangible cultural heritage games. In total, 24 participants took part in the study (13 women, 11 men; aged 18–34), most of whom had limited prior VR experience.

Almost none were directly familiar with *Pahetong*. A small group had at most heard the name but did not know what it referred to ($n = 5$), while the majority had not heard of it at all ($n = 19$).

In addition, six participants who had previously taken part in the co-design workshops (two from each of the VR, design, and culture groups) were recruited for a separate workshop-participant validation session. These workshop participants experienced only the prototype corresponding to their own group, and their data were not included in the main within-subjects questionnaire analyses.

All procedures received institutional review board approval. All participants provided written informed consent before the study and were reminded that they could withdraw at any time without penalty. Those who completed the full session received a 50 RMB (approximately 7.5 USD) honorarium.

4.2.2 Procedure. The three prototypes were presented in a within-subjects design. Each participant experienced all three versions in a single session. The order of versions was fully counterbalanced across the six possible

sequences, so that each ordering pattern was assigned equally often among the 24 participants. We did not disclose the design rationale or intended differences between the three versions to avoid biasing their responses.

Because *Pahetong* rowing involves a non-trivial control technique, we first ran a short controller tutorial in a simplified river-only scene to teach the VR rowing interaction. For each version, participants first received brief instructions on the controls and then played through the game prototype. Immediately after finishing a version, they completed a post-play questionnaire comprising four components:

- Presence: the Igroup Presence Questionnaire (IPQ) [60];
- Motivation and enjoyment: selected subscales from the Intrinsic Motivation Inventory (IMI) [57];
- A self-constructed cultural understanding scale (See Appendix D.1);
- A self-constructed perceived completeness scale (See Appendix D.2).

After all three versions had been played and their corresponding post-play questionnaires completed, participants answered four forced-choice ranking questions. For each question, they ordered the three versions from 1 (best) to 3 (worst) with respect to: (1) how well the version helped them imagine what participating in *Pahetong* is like, (2) which version was most enjoyable to play, (3) which version was easiest to understand how to play, and (4) which version they would most recommend overall. Participants then filled in the Simulator Sickness Questionnaire (SSQ) [33] and took part in a short semi-structured interview (See Appendix E) about their experiences and preferences.

During gameplay, we recorded each participant's total rowing time for the race segment in each version (in seconds) in order to explore whether in-game performance was related to questionnaire measures.

Workshop participant validation. To examine how well the implemented prototypes matched the original co-design intentions, we invited workshop participants back for a validation session. In this session, each workshop participant played the prototype corresponding to their own group (VR, Design, or Culture), completed a brief self-constructed consistency-check questionnaire about how well the prototype reflected their workshop ideas, and responded to follow-up interview questions about perceived alignment and mismatches between the workshop concepts and the implemented game.

4.2.3 Analysis. We first screened questionnaire data for validity using two indicators: (1) consistency between positively and negatively worded items within each scale, and (2) overall questionnaire completion time. Across all 24 returned questionnaires, the mean completion time was 160.63 s (SD = 37.21 s). Three participants showed markedly shorter completion times (87 s, 96 s, and 102 s, respectively) and clear inconsistencies on reverse-coded items. Their inconsistency flags on reverse-coded items reached 15, 10, and 24 instances, respectively. Based on the combination of these patterns, we excluded these three participants from all subsequent quantitative analyses, resulting in a final sample of $N = 21$ for the within-subjects comparison, while still retaining their responses for the overall post-study ranking questions and qualitative interview analysis, since these data were used descriptively to understand preferences and interpretations rather than to support inferential statistical claims.

For the Likert-type questionnaire subscales (IPQ, IMI, Cultural Understanding, and Perceived Completeness), we computed composite scores. Internal consistency of the two self-constructed scales was assessed separately for each version using Cronbach's α, which ranged from .70 to .86 for Cultural Understanding and from .87 to .89 for Perceived Completeness. We then used repeated-measures ANOVA with version (VR, Design, Culture) as a within-subjects factor. Mauchly's tests indicated that the assumption of sphericity was met for all repeated-measures ANOVAs (all $p \geq .068$). Where omnibus tests indicated significant effects of version, we conducted paired t-tests between versions to examine which pairs differed.

For the four ranking questions, we treated ranks as ordinal data and used Friedman tests to assess overall differences between versions, followed by Wilcoxon signed-rank tests for pairwise comparisons. To relate in-game

performance to subjective experience, we computed Pearson correlations between completion time and each questionnaire subscale, both across all versions and separately by version.

Data from the workshop participant validation questionnaires were summarised descriptively, and interview responses were thematically coded to identify perceived consistencies and discrepancies between co-designed concepts and the implemented prototypes.

5 Results

5.1 Findings on Co-design Process (RQ1)

5.1.1 Participation and Roles. Across all three groups, we observed that participation in the initial brainstorming phase was relatively uniform. Most participants contributed as actors by producing keyword sticky notes and composing short gameplay sentences, and then shared these contributions in turn. In contrast, during the subsequent guided convergence and paper-prototyping phases, interaction patterns became more differentiated.

VR group. In the VR group, V1 most consistently took on a director role during convergence and paper prototyping. Beyond proposing mechanics, V1 repeatedly set decision criteria and closed down options by foregrounding feasibility under VR input constraints. For instance, when the group discussed layered actions during continuous paddling, V1 explicitly argued that performing multiple controller operations simultaneously would be impractical, and therefore pushed the team toward simplified mappings and more implementable rule configurations. In these moments, V1 was not only contributing ideas as an actor, but also steering what would count as an acceptable design and moving the group toward a finalisable specification.

V3 alternated between actor and director roles. V3 frequently reframed the design in terms of the intended gameplay experience and competitive feel, and also proposed concrete interaction mappings that could make specific actions workable in VR. At the same time, V3 occasionally helped the group converge by flagging implementation costs and by negotiating trade-offs between "fun" and controllability, thus supporting direction-setting rather than only adding content.

By contrast, V2 and V4 primarily functioned as whisperers. Their contributions were typically shorter and more targeted, such as suggesting incremental rule tweaks or situational mechanics (for example, how village differences might translate into initial attributes). These inputs often enriched the emerging design but rarely redirected the overall trajectory of decisions. Overall, the VR group showed a clear pattern of role differentiation: one strong director (V1) and one mixed actor-director (V3) shaped the final structure, while the remaining participants mainly supported through local refinements. V1 and V3 also reported the two highest VR expertise scores within the group, with raw mean scores of 4.75 and 4.50, respectively (see Appendix B).

Design group. In the design group, roles were more distributed, but similar dynamics appeared around key decisions. In guided convergence, D1 and D4 most often acted as actors or temporary directors: D1 pushed the village-chief fantasy and suggested integrating culturally meaningful elements into the paddles, while D4 turned these into a more systematic skill system. D2, D5 mainly took whisperer roles, clarifying constraints, questioning genre, and cultural references. D3 participated relatively little, taking on a role closer to an audience member, mainly offering evaluative comments rather than contributing new ideas.

In the paper-prototype phase, D1 and D4 led the translation of ideas into layout and rules, deciding how reincarnation, rhythm-guided paddling, fish collection, and skills were represented. D2, D5 filled in interface details and wording, checked consistency with the earlier story, and adjusted parameters in response to concerns about readability, comfort, and cultural grounding. Compared with the VR group, design participants shared actor roles more evenly, yet the process still consolidated around two primary directors who steered key decisions, while the remaining participants mostly took whisperer roles by offering supportive refinements. Although D1 and D4 led key decisions, they did not report the highest self-rated design-experience scores in the group. D1

and D4 reported design raw means of 4.17 and 4.00, while D5 and D3 reported higher scores of 4.83 and 4.50 respectively (Appendix B).

Culture group. In the culture group, three participants (C3, C4, and C6) primarily acted as whisperers: they repeatedly drew on their childhood memories and first-hand experience of *Pahetong*, recounting how harvesting, village celebrations, and bucket races are organised in practice. Although they did not directly talk in terms of game mechanics, their background stories provided the groundwork for later game concepts. C3, C4, and C6 all had direct *Pahetong* experience, and reported cultural-familiarity raw mean scores of 4.40, 3.80, and 2.80, respectively (see Appendix B). For example, C6's remark that "threshing felt like play" was selected as one of the core game ideas and led to the inclusion of a threshing stage in the prototype. A second core concept came from C2. C1 and C2 mainly took on an actor role: they responded to the facilitator's design prompts, translated others' narrated experiences into concrete game structures and scenes, and articulated how specific elements such as the two-stage sequence could be represented in the prototype. C5 mostly positioned themselves as an audience and occasional whisperer, supporting C1 and C2's ideas and adding small contextual details rather than steering the overall direction of the design.

Notably, unlike the VR and design groups, the culture group did not produce a clear participant-level director who orchestrated decisions. Their internal discussions focused more on reconstructing the lived experience of *Pahetong* than on the game structure itself. Process-level directing, such as managing time and moving between topics, was largely carried out by the facilitator, while participants collectively contributed lived experience and design proposals.

5.1.2 Knowledge and Content. Viewed through the Knowledge and Content lens, the sticky-note keywords generated in the third-stage divergent ideation already reveal clear contrasts between the three groups. As shown in Table 2, the VR group contributed twenty-nine notes, the design group thirty-seven, and the cultural group thirty-four. Although they responded to the same design brief, the three groups drew on quite different kinds of heritage- and game-related knowledge.

In the VR group, the largest share of notes related to core game mechanics and to environment and sensory experiences. Participants frequently framed *Pahetong* in terms of exploration, competition, and progression, using keywords such as "exploration", "duel", "competition", "racing", "levels", and "choose track". A comparable number of notes described sensory and environmental qualities of the imagined *Pahetong* experience, including "sun", "heat", "splashing water", "shaking", "waves", "fatigue", "river basin/scene", and "scenery". Interaction and input mechanics were present but less dominant, for example, "modify equipment", "balance", and "falling into the water". Sociocultural meaning appeared mainly as abstract labels such as "folk customs", "appearance", "music", and "social interaction". Overall, the VR group oriented to *Pahetong* primarily as a setting for a competitive and physically engaging VR game, with heritage evoked more as a thematic flavor than as detailed practice.

The design group showed a different knowledge profile. While it also produced many notes on interaction and input mechanics, this group did so by explicitly drawing on reference games and technical terms. Participants named titles such as "Mario Party", "Mario Kart", "Subway Surfers", "Temple Run", and "Ring Fit Adventure" alongside terms like "parkour", "coins", "skills", "co-op", "competitive cooperation", "hand movements", "paddling", and "multimodal interaction". Environment and sensory experience appeared in references to "visual style", "roller-coaster-like bodily experience", and "seasickness". Compared with the VR group, the design group articulated a richer layer of sociocultural meaning, including "family affection", "friendship", "traditional story plot", and "dialect", yet these references remained relatively abstract. In this group, *Pahetong* was often imagined as something to be embedded within familiar game genres and interaction paradigms, while simultaneously carrying emotional and narrative hooks.

The cultural group showed a markedly different content profile from the other two groups. Notably, none of their notes referred to core game mechanics or to reference games. Instead, their contributions clustered around

concrete scenes of the heritage practice, bodily techniques and remembered situations. In terms of environment and sensory experience, participants wrote about "casting the net", "ducks", "water sounds", "small boats", "straw hats", "fishing nets", "choking on water", and "cheering", sketching a vivid soundscape and riverside setting. Interaction and input were described through practical actions rather than abstract control schemes, for example, "the bucket easily wobbles, you must master balance", "crashing into the reeds", and "some people sit to paddle, some kneel". A distinctive category in this group concerned concrete scenes of the heritage practice that linked *Pahetong* to agricultural work. Participants repeatedly referred to threshing rice, for instance, "the more you thresh, the more work points you earn", "do not let the grain fall on the ground", and "threshing is like playing". Sociocultural meaning was expressed through explicit reflections and aspirations, including "extremely lively, the most unforgettable childhood memory", "help people recover their memories" and "I hope the village will organise this activity again for our generation born in the 1960s". Taken together, the cultural group treated *Pahetong* not as a game concept but as an intertwined experience of paddling and threshing, bodily skill, agricultural labour, and collective memory.

Table 2. Distribution of sticky-note keyword categories across groups

Category	VR group	Design group	Cultural group
Core game mechanics	11	7	0
Reference games and technical terms	0	9	0
Environment and sensory experience	9	4	8
Sociocultural meaning	3	4	3
Interaction and input mechanics	6	13	15
Concrete scenes of the heritage practice	0	0	8
Total	29	37	34

5.1.3 Interaction and Decision-Making. Across the three sessions, interaction and decision-making unfolded within the same scaffold. Each workshop moved from brainstorming with keyword cards and sentence making, to voting on core game concepts, then defining the Experience, Play and Rules dimensions, and finally summarising the concept and producing a paper prototype. As shown in Figure 5, all three groups devoted the largest share of time to early divergent work and to negotiating Experience and Play. The brainstorming and sentence sharing phases were strongly facilitator led and consisted mainly of structured turn taking, where participants responded in sequence and the facilitator paraphrased. More spontaneous, participant initiated discussion emerged only after the first vote, when some ideas had to be dropped, and again when abstract ideas were translated into concrete mechanics and prototype elements.

Across the three groups, the Experience, Play and Rules phases differed in how initiative and control were distributed. In the VR group, the Experience phase followed a clear pattern of facilitator questions, individual sharing and a short vote, after which participants could finalise the Play dimension largely through their own answers, with the facilitator summarising and others adding clarifications. Rules were negotiated in open discussions. When input methods were raised, V3 proposed that "for simplicity it is better to use the trigger", and V1 extended this by suggesting that all skills and actions be simplified to automatic pickup; the group accepted these proposals without further alternatives. In the design group, the Experience and Play phases gave rise to more complex systemic ideas and required more time to reach agreement. During the Play phase, participants proposed numerous ideas about game goals. For example, D4 suggested adding affinity systems, richer NPC interactions and even alliance tournaments between villages, likening this to MOBA-style team competitions.

This drifted away from the three core concepts agreed earlier, and the facilitator responded by revisiting those concepts and steering the group back to the main line of discussion. In the subsequent exchange, D1, D3, D4 and D5 engaged in a focused debate about what rhythmic structure paddling should follow and, under the facilitator's guidance, eventually converged on a shared solution.

The culture group showed a different interactional pattern and a more directive role for the facilitator. In the Experience phase, several participants initially misinterpreted the question "who is the player" as asking about the target audience, and C3 described her preference for watching others paddle rather than inhabiting a role. The facilitator had to restate the question, explain the notion of an in game role, and lead participants toward a common answer. Elders such as C3, C4 and C6 shared detailed memories of *Pahetong*, and under the facilitator's guidance, C1 and C2 reformulated these recollections into experiential statements that could be more directly translated into game concepts. In the Play phase, participants continued to describe the goals of real competitions in objective historical terms, for example, C6's explanation that in the past pork and rice were given as prizes. The facilitator encouraged the group to reframe such accounts in more game-like terms, for instance, as "paddling faster to obtain a higher ranking and rewards". Here, decision-making rested more clearly with the facilitator as mediator between lived experience and the Experience, Play, and Rules categories used in the other workshops.

Finally, the paper prototype phase was used in distinct ways. In the VR and design groups, this phase supported refinement of Rules, including stamina and speed values, collision outcomes, respawn conditions and the smallest playable loop, as participants manipulated tokens and agreed on details through rapid situated negotiation. In the culture group, participants mainly used paper, pens and simple props to model in game tools and enact scenes such as threshing and hitting field mice. Rule related points were briefly summarised rather than deepened, and the emphasis remained on representing the setting and activities rather than extending the underlying game logic.

5.2 Findings on Co-design Outcomes (RQ2)

5.2.1 Workshop Participant Validation. Six workshop participants (two from each of the VR, design, and culture groups) completed a four-item consistency check questionnaire after playing the prototype corresponding to their own workshop. Items were rated on a 7-point Likert scale; the negatively worded item ("some important elements were missing from this prototype") was reverse-scored so that higher values consistently indicated stronger perceived alignment between the prototype and the original workshop ideas.

Descriptively, the overall consistency score was high, with a mean of 6.42 (SD = 0.42) out of 7. Group means showed a similar pattern (VR: 6.63; Design: 6.13; Culture: 6.50), all above 6 on average. Item-level means also clustered between 6.0 and 7.0, indicating that participants felt the prototypes broadly preserved the core game concepts, mechanics, and representation of *Pahetong* developed in their workshops. Given the very small sample, we treat this as a descriptive indication of perceived consistency rather than a robust statistical result.

Qualitative feedback from the same six workshop participants further clarified how they perceived the relationship between their co-designed concepts and the implemented prototypes. Overall, participants across all three groups reported that the core elements discussed in the workshops were clearly reflected in the games and affirmed our interpretation of their design ideas. Participants in the culture group felt that the prototype closely matched their real experience of first harvesting rice and then taking part in *Pahetong*. At the same time, participants noted that certain numerical parameters and in-game effects could be further refined; for example, members of the design group suggested that the rhythm of the rowing chants could be made more salient. Taken together, they framed these shortcomings as consequences of technical and time constraints rather than conceptual misunderstanding, and indicated that such gaps were understandable and acceptable.

5.2.2 Quantitative results. As detailed in Section 4.2.3, three participants were excluded due to inconsistent responses and abnormally short completion times. Questionnaire data from $N = 21$ participants who completed

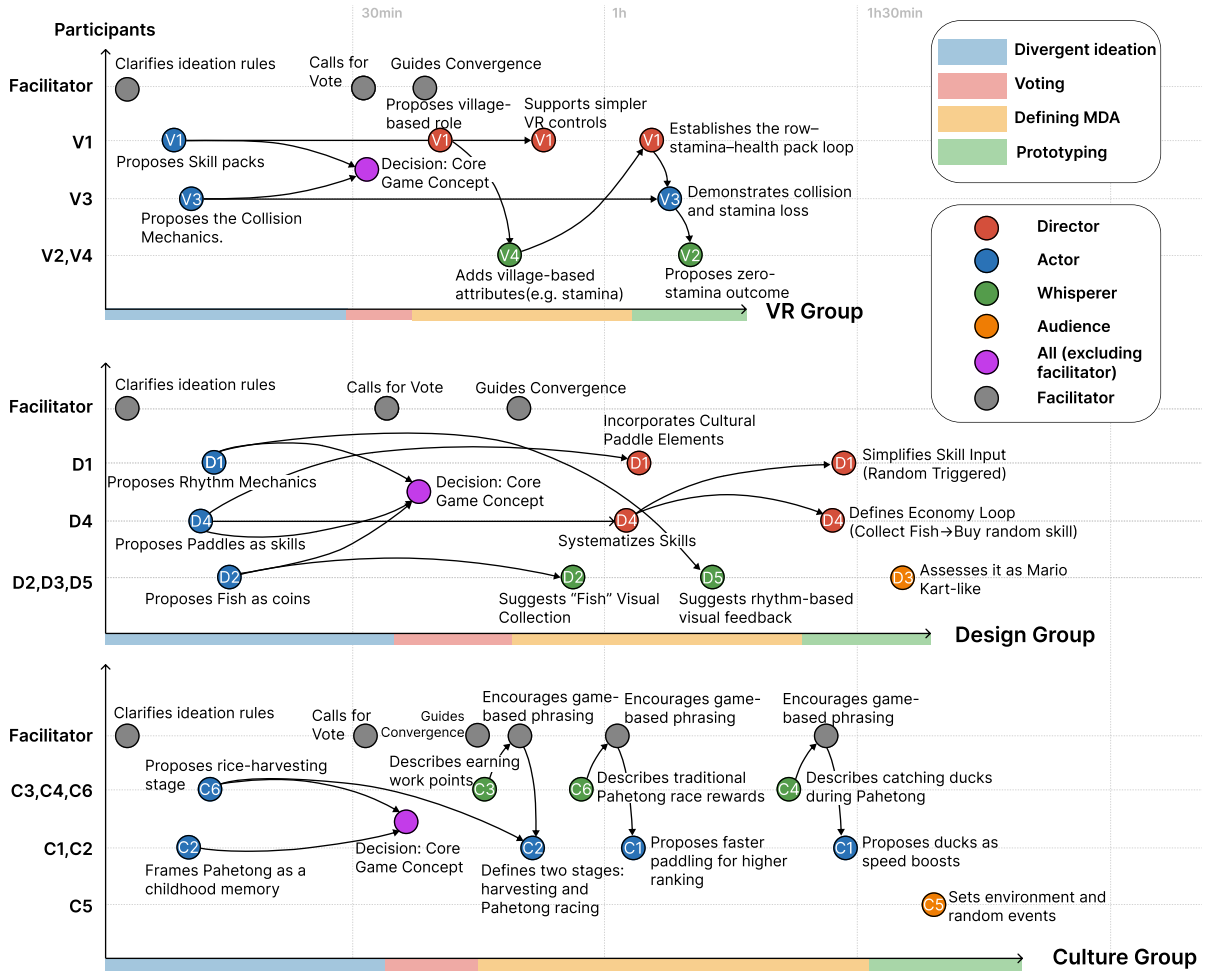

Fig. 5. Distribution of participation roles and key design moves across three stakeholder-configured co-design workshops over time. The horizontal axis shows workshop phases from divergent ideation to voting, defining MDA dimensions, and paper prototyping, while the colored bars indicate workshop phases, and colored dot markers indicate director, actor, whisperer, audience, and facilitator roles in each group. Annotated callouts summarise representative contributions, such as proposing collision mechanics and stamina–health pack loops in the VR group, incorporating culturally embedded paddle elements and fish-based economies in the design group, and defining harvesting–Pahetong dual stages and work-point systems in the culture group. The diagram highlights how structured brainstorming transitioned into differentiated role dynamics and concrete rule formation across sessions.

all three versions were entered into a within-subjects analysis. Table 3 summarises means and repeated-measures ANOVA results, and Figure 6 shows the distributions for the subscales that yielded significant effects. The p-values reported for the repeated-measures ANOVAs and post-hoc paired comparisons were not adjusted for multiple comparisons and should therefore be interpreted as exploratory.

Table 3. Descriptive statistics and repeated-measures ANOVA results for all questionnaire subscales ($N = 21$; $^{*}p < .05$; $^{**}p < .01$).

Measure	VR (M ± SD)	Design (M ± SD)	Culture (M ± SD)	$F(2, 40)$	p
IPQ (presence)					
Sense of being there	5.76 ± 0.89	5.95 ± 0.81	5.95 ± 0.81	0.84	0.441
Spatial presence*	5.48 ± 0.87	5.71 ± 0.73	5.66 ± 0.78	4.05	**0.025**
Involvement	3.88 ± 1.03	3.92 ± 1.30	3.76 ± 0.86	0.24	0.791
Realism	4.29 ± 1.08	4.54 ± 1.14	4.50 ± 1.22	0.93	0.403
IMI (motivation)					
Interest / enjoyment	5.39 ± 1.13	5.82 ± 0.81	5.52 ± 1.06	2.12	0.133
Perceived competence*	4.18 ± 1.57	5.33 ± 0.98	4.71 ± 1.36	4.39	**0.019**
Effort / importance	5.24 ± 0.97	5.30 ± 1.01	5.31 ± 0.94	0.12	0.892
Pressure / tension	3.08 ± 1.19	2.86 ± 1.25	3.06 ± 1.45	0.62	0.542
Value / usefulness*	5.65 ± 1.24	6.10 ± 0.75	5.96 ± 0.87	3.62	**0.036**
Cultural understanding					
Cultural understanding score	5.66 ± 1.05	5.98 ± 0.81	5.86 ± 0.72	2.21	0.122
Perceived completeness					
Completeness score**	5.19 ± 1.24	5.80 ± 0.97	5.37 ± 1.19	7.30	**0.002**

For presence (IPQ), only spatial presence showed a significant main effect of version, $F(2, 40) = 4.05$, $p = .025$. Post-hoc paired t-tests indicated that the design version elicited higher spatial presence than the VR version (mean difference = 0.23, $p = .012$, Cohen's $d_z = 0.60$), while the culture version did not differ significantly from either of the other two. Sense of being there, involvement, and realism did not differ reliably across versions.

For motivation (IMI), perceived competence differed significantly between versions, $F(2, 40) = 4.39$, $p = .019$. Participants felt more competent in the design version than in the VR version ($p = .008$, $d_z = 0.65$); the culture version lay in between and did not differ significantly from either. Interest/enjoyment, effort/importance, and pressure/tension showed no significant main effects. Perceived value/usefulness showed a significant omnibus effect, $F(2, 40) = 3.62$, $p = .036$, driven by higher ratings for the design version than for the VR version in the pairwise comparison ($p = .025$).

The cultural-understanding composite did not show a significant main effect of version, even though all three prototypes were rated relatively high on this dimension (means between 5.66 and 5.98 on a 1–7 scale). By contrast, the perceived completeness scale showed the clearest differences, $F(2, 40) = 7.30$, $p = .002$. Post-hoc tests indicated that the design version was perceived as more complete and polished than both the VR version ($p = .001$, $d_z = 0.86$) and the culture version ($p = .011$, $d_z = 0.61$), while the VR and culture versions did not differ significantly from one another. Together, these results suggest that participants generally experienced similar levels of presence and cultural understanding across versions, but saw the design prototype as somewhat more spatially involving, easier to play competently, and more finished as a game.

We also explored how in-game performance related to subjective experience. For each version, we recorded the time (in seconds) required to paddle from the start to the finish line. In the pooled analysis across all three versions (63 observations), completion time showed a moderate negative association with perceived competence ($r = -0.54$, $p < .001$) and a small positive association with pressure/tension ($r = 0.29$, $p = .020$), indicating that

faster players tended to report higher competence, whereas slower players tended to report higher pressure. Other subscales, including spatial presence, cultural-understanding scores, and perceived completeness, showed no notable pooled associations with completion time (all $|r| < 0.30$, all $p > .10$). Because each participant contributed three repeated observations, the pooled observations were not statistically independent, and these correlations should therefore be interpreted as exploratory.

Version-specific analyses showed that completion time was negatively associated with perceived competence in the VR version ($r = -0.64$, $p = .002$) and the Design version ($r = -0.52$, $p = .015$), but not in the Culture version ($r = -0.35$, $p = .121$). No other version-specific correlations were statistically significant.

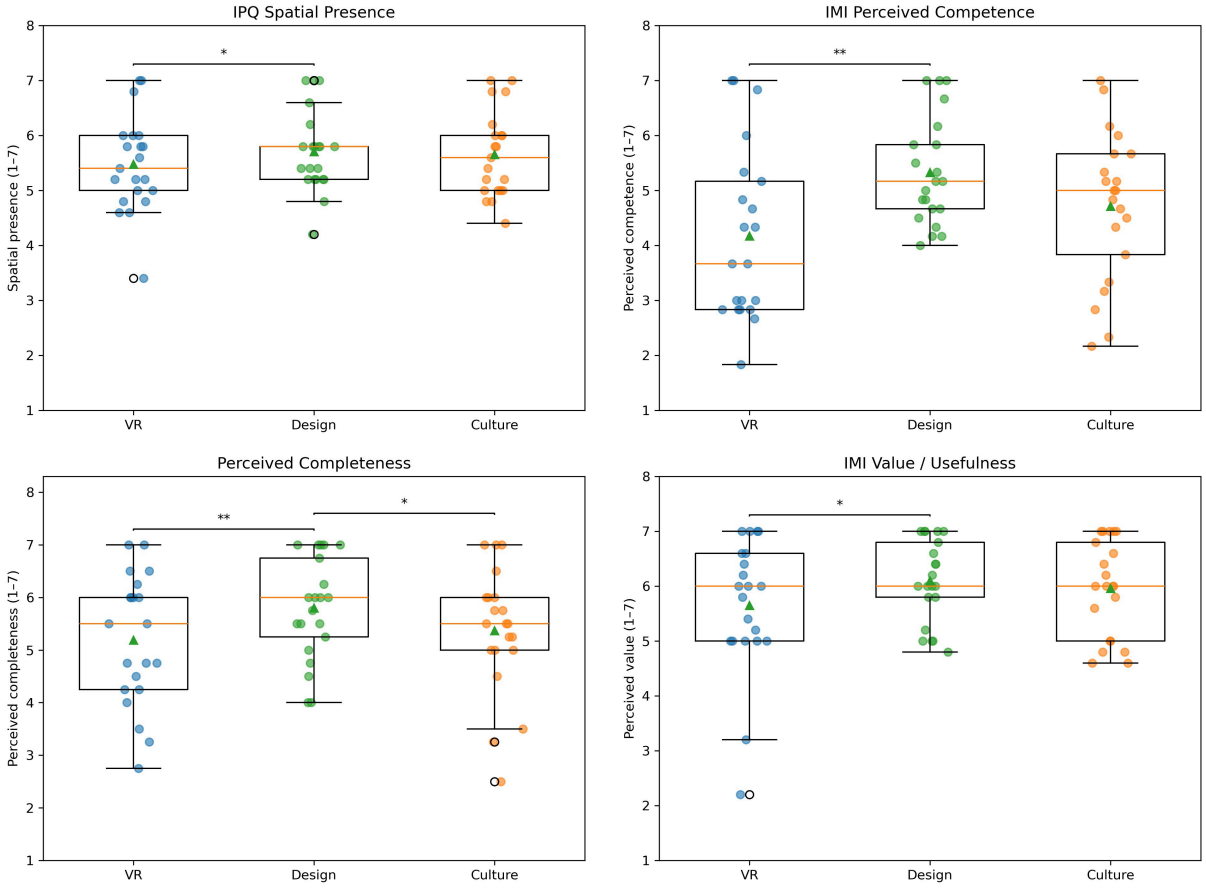

Fig. 6. Boxplots of questionnaire scores for the three *Pahetong* VR prototypes (VR, Design, Culture). Each panel shows one subscale (spatial presence, perceived competence, perceived completeness, and perceived value/usefulness), with scores on a 1–7 Likert scale. Boxes indicate the interquartile range with the median as the central line; whiskers extend to 1.5 IQR, and jittered points represent individual participant scores (N = 21).

As detailed in Section 4.2.2, participants provided forced rankings of the three versions on four criteria. Descriptively, the design version tended to receive the best ranks for conveying what *Pahetong* is like (mean rank = 1.67, SD = 0.76, 12 first-place votes), followed by the culture version (mean = 2.04, SD = 0.81, 7 first-place

votes) and the VR version (mean = 2.29, SD = 0.81, 5 first-place votes). For enjoyment, the culture version was most often ranked first (mean = 1.75, SD = 0.94, 14 first-place votes), with design (means = 2.04, SD= 0.55, 3 first-place votes) and VR (means = 2.20, SD= 0.88, 7 first-place votes) trailing. For ease of understanding, VR was most frequently ranked as easiest (mean = 1.67, SD = 0.76, 12 first-place votes), followed by design (mean = 2.08, SD = 0.78, 6 first-place votes) and culture (mean = 2.25, SD = 0.85, 6 first-place votes). For the overall recommendation question, first-place votes were more evenly distributed across versions (VR: 9, design: 6, culture: 9; mean ranks = 1.91, 2.00, and 2.08, respectively).

Friedman tests did not show statistically significant overall differences between versions on any of the four ranking questions (conveying *Pahetong*: $\chi^2(2) = 4.75$, $p = .093$; enjoyment: $\chi^2(2) = 2.58$, $p = .275$; ease of understanding: $\chi^2(2) = 4.33$, $p = .115$; overall recommendation: $\chi^2(2) = 0.33$, $p = .846$). We therefore interpret the ranking patterns as descriptive indications of preference rather than as conclusive evidence of superiority, and return to them in Section 5.2.3 when discussing participants' qualitative explanations.

SSQ scores were generally very low across participants ($N = 24$). Mean nausea, oculomotor and disorientation subscores were 2.27 ($SD = 3.76$), 2.39 ($SD = 3.75$) and 3.65 ($SD = 7.13$), respectively. The weighted total SSQ score averaged 1.02 ($SD = 1.66$), with observed values ranging from 0.00 to 6.31. Taken together, these results indicate minimal simulator sickness symptoms in our sample.

5.2.3 *Qualitative Results.*

Version preferences and perceived trade-offs. Interview data helped explain how participants made sense of the three versions when ranking them. Consistent with the quantitative rankings, 12 participants described the design version as best at evoking the overall scene of *Pahetong*, emphasising elements such as the rowing chants, fish collection, and visual composition of the river environment as helping them "picture how the event takes place". For example, P2 remarked that "the dialect rowing chants made the experience much stronger and made the whole event atmosphere feel very rich". P19 similarly noted that "the work chants together with the fish made me immediately feel that this was a local folk event".

Fourteen participants ranked the culture version as the most enjoyable. They typically explained this by saying that it offered a richer variety of scenes and activity patterns, rather than focusing solely on paddling for speed. In addition, P4 commented that "this version made me feel the joy of a good harvest." Seven participants ranked the VR version as the most enjoyable; they valued its competitive qualities and felt that the collision mechanics created a sense of social, competitive fun.

Twelve participants identified the VR version as the easiest to understand, often describing it as the most "game-like". At the same time, several felt that its interface and skill elements "felt more like a generic commercial game and less specifically tied to *Pahetong*". As P14 put it, "the VR version is the clearest as a game, but it also feels most like a normal game you could download anywhere, not especially about *Pahetong*." Taken together, these accounts suggest that participants were navigating a trade-off: the VR version was experienced as clearer and more competitive, the design version as better balancing game structure with cultural cues, and the culture version as distinctive but more uneven in terms of clarity and pacing.

Understanding Pahetong as a cultural practice. Across interviews, participants were generally able to form a basic picture of *Pahetong* after playing the three versions, describing it as a traditional collective rowing event linked to harvest-time celebrations in a local village context. A small number of participants, such as P7, explicitly inferred from the culture version that rice harvesting was a necessary precondition for *Pahetong*. More commonly, before we provided any background explanation, participants understood *Pahetong* in broader terms as a harvest-related folk event without grasping the full sequence of practices. They reported that this impression mainly arose from the rice-field scenes and duck elements in the culture version, the work chants and fish in the design version, and the crowds of onlookers in both the VR and culture versions. For example, P17 described

the culture version as feeling "very authentic, with all the ducks as props and rice paddies that really match the local setting", and several participants (P4, P14, P19, P21, P24) noted that the presence of spectators along the riverbank contributed to a sense of lively celebration.

At the same time, before our brief introduction to *Pahetong*, participants frequently reported being unsure how the first rice-harvesting stage in the culture version was connected to the later bucket race. After the explanation, most were able to reconcile what they had done in the game with the idea of *Pahetong* as a harvest festival involving rowing. However, some participants (P2, P13, P16, P21, P23) still reported that they could not clearly understand the historical origin or functional role of the wooden bucket from the game alone, for instance, its use in collecting and transporting grain. Taken together, these accounts suggest that, in the context of a VR game, the three prototypes successfully communicated *Pahetong* as a harvest-time folk rowing event with characteristic symbols and atmosphere, yet the historical trajectory of the practice and the specific role of the bucket remained largely implicit and required additional narrative scaffolding for most players to fully grasp.

Game-like feel, interaction, and control. Participants also reflected on how each version felt as a game and how interaction design shaped their experience. The VR version was perceived as the most intuitive to play, as its use of skill packs and a health bar mirrored mainstream game conventions, but participants also felt that this made it closer to a generic commercial game than to a specifically *Pahetong*-themed experience. In contrast, the culture version was often characterised as closer to a realistic simulation. Participants commented that its scenes felt like a direct representation of local life and rituals (P2, P3, P13, P15, P16, P19, P21). They appreciated this cultural fidelity, yet some also felt that the presentation was overly literal and lacked a strong sense of gameplay (P3, P16, P19). Many participants expressed particular appreciation for the work chants in the design version, noting that they helped them feel more immersed in the rowing scene and "row with more energy" (P2, P4, P8, P9, P19, P20).

In the design and culture versions, several participants remarked that the additional button-based interaction was rarely used in practice (P1, P9, P10, P13, P21, P22, P23). They reported that pressing a trigger to activate an on-screen button while paddling or harvesting felt discontinuous, because their attention was focused on the core rowing or harvesting action, and they did not spontaneously think to perform an extra button press.

Across all three versions, almost all participants reported that using the controllers to simulate *Pahetong* rowing was something that needed to be learned and could feel difficult. Despite the presence of a tutorial, they still found themselves spinning in place or colliding with the riverbank during play. Although many participants appreciated the competitive aspects of the VR game, some explicitly noted that the collision mechanics and health bar made the controls feel more demanding (P13, P16).

6 Discussion

6.1 Structuring Participation and Role Dynamics in Co-design

In our workshops, role dynamics unfolded on the foundation of participant-authored sentences: the keyword task ensured that everyone first contributed concrete materials, and subsequent discussion built from these shared starting points, allowing actor, director, and whisperer roles to crystallise through negotiation and elaboration rather than pre-assigned authority (See 3.2.3). Compared with co-creative processes grounded in tangible visual artefacts, such as Ppali et al.'s art-making workflow, where participants externalised ideas through making and then translated them into immersive environments [55], our sentence-based elicitation operated at a more abstract level. This abstraction is efficient for producing diverse concepts, but it also makes later convergence dependent on participants' ability to operationalise ideas into rules and interactions. In this respect, prior work suggests that facilitation and mediation strategies become especially important when stakeholders differ in design literacy or cannot comfortably co-participate in the same setting, because the pathway from expressed ideas to implementable decisions requires deliberate structuring and reconciliation [34, 53]. A related contrast appears in Liaqat et al.'s participatory work, where collaborative narration scaffolds participation through sustained

storytelling and shared meaning-making over time, whereas our workflow front-loads authorship and then relies on the group to negotiate which concepts become the "game" [38, 39]. Finally, Vayanou et al. show that distributing a mediator role can actively shape group dialogue and perspective sharing in cultural group games, highlighting an alternative approach where role structures are explicitly designed into interaction rather than emerging only through conversation [70].

Beyond task structure, our comparative analysis of the three groups shows that different forms of expertise shaped how influence and direction accumulated during convergence. Although all groups began from an equalised authorship stage, clearer participant-level leadership emerged in the VR and design groups, whereas the culture group relied more on facilitator coordination as discussion stayed closer to reconstructing lived experience. This pattern complicates the assumption that domain expertise straightforwardly translates into design leadership. Ko et al. show how community-embedded, design-novice facilitators enact participatory values by creating respectful, non-hierarchical conditions and making space for participants' knowledge to shape the process and outcomes [34]. In our case, cultural knowledge most strongly expanded the interpretive palette of what could be represented, while the momentum of turning ideas into concrete rules tended to be carried by those with stronger design fluency or technical confidence. Bratteteig and Wagner similarly emphasise that participation is reflected in whether participants' contributions shape design choices, rather than simply in being invited to contribute [8]. This pattern is also reflected in the participant-level expertise profiles, which suggest that self-rated expertise and workshop influence may be related, but were not interchangeable. While the strongest directors in the VR group also reported the highest VR expertise, this pattern did not hold as clearly in the design group. In the culture group, participants with direct *Pahetong* experience contributed many of the lived memories and contextual details that informed the design, but did not necessarily take leading roles in translating these experiences into game structures.

6.2 From Cultural Experience to Legible Design

A central finding of our study is that *cultural authenticity did not automatically translate into interpretability* (See 5.2). Participants in the culture workshop, who had first-hand experience of *Pahetong*, perceived their prototype as highly consistent with real practice. However, newcomers without prior knowledge struggled to understand the first level and its relationship to the broader activity. While elements such as rice fields, threshing sequences, and work points carried clear meaning for insiders, they functioned only as implicit cues for others. This gap suggests that experiential fidelity alone is insufficient to ensure legible design.

This tension can be understood through the lens of the "curse of knowledge" [11], which describes how experts may underestimate the interpretive challenges faced by novices once knowledge becomes internalized. Participatory HCI research cautions against assuming that inviting experiential experts automatically leads to shared understanding, as participation is shaped by facilitation and the distribution of control in practice [71]. Rather, Participatory design has been framed as a "third space" in which heterogeneous forms of knowledge must be negotiated and translated [49]. In our case, the culture group effectively articulated lived memories, yet the resulting design implicitly relied on background knowledge that audiences failed to understand.

At a more structural level, this issue reflects the difficulty of externalizing tacit knowledge. Intangible cultural heritage practices are often transmitted through embodied participation and oral tradition, which resist straightforward codification [26]. The culture workshop successfully accumulated and expressed insider experience, but diffusion toward non-expert users remained limited. By contrast, the design group introduced clearer interpretive framing through rhythmic chants and more structured task progression, which functioned as scaffolding. This supports prior findings that contextual storytelling and explicit framing mechanisms enhance interpretability in VR heritage systems [79].

Heritage game scholarship further reinforces that accuracy does not guarantee comprehension. Mustonen et al. [50] describe cultural game development as an ongoing negotiation between fidelity and playability, requiring deliberate mediation rather than direct reproduction. Systematic reviews of VR heritage applications similarly highlight the role of structured interaction and guidance in supporting user understanding [67, 83]. Our findings extend this literature by demonstrating that even when cultural insiders directly participate in co-design, legibility must still be actively constructed. Moving from cultural experience to legible design, therefore, requires not the replication of lived details but the transformation of tacit practice into interpretable structures. This implication should be understood in relation to the specific affordances of *Pahetong*. As an embodied, activity-oriented practice organized around paddling, racing, rhythm, and spatial movement, *Pahetong* is relatively well suited to translation into VR game mechanics. Other forms of ICH may not map as directly onto playable action, but related work suggests that they can still be transformed into legible digital experiences through different mediation strategies. Studies of Hua'er and historical storytelling show how oral performance and historical events can be reorganized through avatars, participatory performance, narrative sequencing, guided explanation, and task-based exploration [16, 41]. Cho et al.'s shamanism-inspired conversational AI further suggests that ritual meaning can be translated through atmosphere, symbolic objects, voice, constrained control, and personalized interaction[15]. Together, these works resonate with our finding that cultural experience does not become legible simply by being represented. Across embodied practices, oral performance, historical storytelling, and ritual-like digital experiences, tacit or context-dependent cultural knowledge must be actively mediated into forms that users can perform, follow, feel, or interpret. In this process, design practitioners play a critical role as mediators who translate embodied heritage knowledge into encounters that remain meaningful to insiders while becoming legible to newcomers.

6.3 Translating Co-design into Playable VR Interaction

Our results show that culturally meaningful workshop ideas do not automatically become playable VR interactions once implemented. Participants generally found the VR version easiest to pick up, whereas the design and culture versions introduced an extra button-based action that was often ignored during rowing and harvesting. This highlights a practical constraint in co-design for interactive media: beyond deciding what to represent, teams must also decide how actions can be enacted under VR input demands and attentional load. This aligns with participatory design accounts that frame outcomes as negotiated in a "third space" between stakeholder knowledge and professional design practice [49], and with recent CHI reflections cautioning that participation alone does not guarantee implementable, legible outcomes [34].

The constraint is especially salient in VR, where embodied controllers and interaction mappings can render conceptually meaningful mechanics awkward or cognitively costly during continuous action. Prior VR game research shows that tutorial and instruction design for controllers can measurably affect learnability, performance, and player experience under varying control demands [32]. In our study, the frequently unused button interaction illustrates how even small additional input requirements can be deprioritised when attention is already absorbed by sensorimotor coordination.

This translation challenge is amplified for intangible cultural heritage, where understanding is grounded in tacit, embodied skill. *Pahetong* is not only cultural content to communicate, but also a technique to enact, and the skill demands reappear in VR as a learning curve, even with tutorials. Consistent with this, faster completion times aligned with higher perceived competence, whereas slower times co-occurred with greater pressure/tension, echoing HCI accounts that stress the difficulty of externalising tacit knowledge through explicit instructions alone [64].

Finally, interaction rules can reshape cultural experience even when assets remain constant. In the VR version, collision-based competition increased enjoyment but also raised control difficulty, potentially shifting attention

toward steering and recovery rather than cultural cues. Related heritage work likewise shows that interaction components and system complexity shape how users allocate attention and make meaning in digital heritage experiences [62]. Taken together, translating co-design concepts into playable heritage VR interaction requires balancing input constraints, interaction difficulty, and cultural legibility.

6.4 Design Implications

Our findings suggest three design implications for researchers, designers, and cultural institutions developing VR games for intangible heritage. First, co-design workshops should separate distributed contribution from convergence. In our workshops, the keyword and sentence-making tasks helped establish distributed contribution by ensuring that all participants first produced concrete cultural memories, sensory details, interaction references, or game ideas. Later convergence, however, depended on who could translate these materials into rules, interactions, and feasible prototypes. Designers can therefore structure workshops in two stages: first, use low-barrier prompts to support distributed contribution from all participants; second, make the translation process explicit by mapping selected ideas onto concrete game-design categories, such as player goals, required actions, feedback mechanisms, and rule conditions. This staged structure can help preserve broad participation while acknowledging that different participants may contribute differently during implementation-oriented decisions. In this sense, it extends calls for more explicit democratic engagement in participatory design by supporting both distributed contribution and clearer co-decision-making around technology options and design choices [34].

Second, cultural authenticity should be treated as a starting point rather than an endpoint. The culture-group prototype was perceived by insiders as closely aligned with their lived *Pahetong* experiences, yet many newcomers struggled to understand how harvesting, work points, and bucket racing were connected. Designers could therefore avoid assuming that accurate cultural details will be self-explanatory. Instead, culturally meaningful elements should be paired with interpretive scaffolding, such as staged task progression, contextual explanation, onboarding moments, rhythmic or sensory cues that can shape engagement and embodied experience in VR games [80], and narrative framing that explains why an action matters culturally, not only how it is performed. This is especially important when tacit knowledge is familiar to insiders but opaque to first-time players.

Third, co-designed concepts should be treated as materials for translation rather than ready-made mechanics. Before implementation, each proposed cultural element should be checked against three questions: (1) what cultural meaning it carries, (2) what player action it requires, and (3) whether that action remains learnable under VR input constraints. In our prototypes, culturally meaningful additions such as button-based tool use were often ignored during continuous paddling or harvesting, suggesting that meaningful mechanics can still fail if they compete with other embodied control demands. This check can help designers balance cultural specificity, playability, and player comprehension before committing workshop ideas to implementation.

6.5 Limitations and Future Work

This work has several limitations. First, the co-design process was conducted with small groups of four to six participants and relied on a single 90-minute workshop for each group. While this format supported rapid ideation, it constrained the representativeness and depth of the resulting proposals. Participatory design research has long argued that robust co-creation typically requires sustained engagement, iterative negotiation, and mutual learning across multiple sessions, rather than being achievable through one-off workshops [8, 49]. In cultural heritage and community-oriented contexts, longer-term participatory engagements further enable stakeholders to co-construct shared vocabularies and progressively refine prototypes over time [1, 46].

Second, although we conducted a within-subjects quantitative evaluation, the final sample size was modest, and the stakeholder configurations were based on participants' self-reported levels of cultural familiarity, design

experience, and VR literacy. In addition, one Cultural Understanding item assessed interest in learning more about *Pahetong*, potentially reducing the scale's sensitivity to version differences. Accordingly, we interpret the quantitative findings as contextual, trend-level evidence that helps triangulate the qualitative accounts, rather than as broadly generalizable effects.

Another limitation concerns how stakeholder configurations should be interpreted. Our workshops were designed to compare expertise-oriented configurations, rather than to produce representative samples of cultural practitioners, designers, or VR users. This design made it possible to observe how different forms of knowledge entered the same workshop structure, but it also means that the findings should not be read as claims about stakeholder categories as a whole. This issue is especially important for cultural representation: the culture group contributed situated memories and direct *Pahetong* experiences, rather than a comprehensive or authoritative account of the broader community. Similarly, the design- and VR-oriented groups were not fully separable categories, since several participants reported substantial experience across both dimensions. These overlaps are not only limitations, but also reflect the interdisciplinary nature of heritage VR design. Facilitator influence should also be understood in this context: the same facilitation protocol supported comparability across workshops, but the facilitator necessarily shaped the discussion as it moved from cultural narration to design convergence.

Building on these limitations, future work should move from short, isolated workshops toward a more sustained heritage VR co-design process that connects broader participation, staged translation, and longer-term evaluation. First, future studies should involve larger and more internally varied stakeholder groups, including multiple local voices across generations, practitioner roles, and levels of cultural involvement, so that researchers can examine not only how cultural accounts are contributed, but also how they are selected, prioritized, or contested during design. Second, the co-design process itself could be organized as a staged participatory pipeline rather than a single-session workshop: culturally experienced practitioners could establish context, key practices, and interpretive priorities; design-experienced participants could help translate tacit knowledge into metaphors, rhythms, narrative frames, and legible interaction structures; VR-literate contributors could validate feasibility under embodied input constraints and controller mappings; and broader community members could iteratively test and refine interpretive clarity and usability. Such a staged approach would make the translation from cultural knowledge to playable interaction more explicit, while allowing participants to review how their contributions are transformed into prototype decisions. It would also make facilitator mediation more visible, for example, by documenting facilitator interventions or adding follow-up validation sessions in which participants reflect on how cultural meanings were preserved, altered, or omitted. Finally, future work should evaluate heritage VR experiences beyond immediate post-play responses through longitudinal and in-situ studies, such as repeated play sessions or deployments in museums and local cultural events, to examine how cultural understanding, memory, and interpretation develop over time [79]. Together, these directions align with arguments that participatory outcomes emerge through ongoing negotiation and distributed expertise rather than direct transfer from stakeholder ideas to implementable artefacts [34, 49], and with cultural heritage design work that emphasises sustained stakeholder engagement across the production lifecycle [10].

7 Conclusion

This work examines how intangible cultural heritage can be translated into playable VR through co-design, using *Pahetong* as a case study. Across three co-design workshops and the resulting prototypes, we articulate how culturally grounded concepts were translated into VR gameplay, and how the implemented prototypes in turn supported players' understanding of *Pahetong* and shaped their cultural experience in play. Our mixed-method evaluation indicates that role dynamics during the co-design process, together with the embodied input demands of VR, strongly influenced which ideas ultimately became operational mechanics and how intelligible those mechanics felt during gameplay. Collectively, these findings contribute an empirically grounded account of how

participation structure and role differentiation shape co-design outcomes, and offer actionable design guidance for cultural heritage digital experiences.

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References

[1] Gabriela Avram, Luigina Ciolfi, and Laura Maye. 2020. Creating tangible interactions with cultural heritage: lessons learned from a large scale, long term co-design project. *CoDesign* 16, 3 (July 2020), 251–266. doi:10.1080/15710882.2019.1596288

[2] Amid Ayobi, Katarzyna Stawarz, Dmitri Katz, Paul Marshall, Taku Yamagata, Raul Santos-Rodriguez, Peter Flach, and Aisling Ann O'Kane. 2021. Co-Designing Personal Health? Multidisciplinary Benefits and Challenges in Informing Diabetes Self-Care Technologies. *Proc. ACM Hum.-Comput. Interact.* 5, CSCW2, Article 457 (Oct. 2021), 26 pages. doi:10.1145/3479601

[3] Barbara Balbi and Alessandra Marasco. 2021. Co-designing Cultural Heritage Experiences for All with Virtual Reality: a Scenario-Based Design approach. *Umanistica Digitale* 11 (2021), 71–98. doi:10.6092/issn.2532-8816/13686

[4] Gabriel Barata, Sandra Gama, Joaquim A.P. Jorge, and Daniel J.V. Gonçalves. 2014. Relating gaming habits with student performance in a gamified learning experience. In *Proceedings of the First ACM SIGCHI Annual Symposium on Computer-Human Interaction in Play* (Toronto, Ontario, Canada) *(CHI PLAY '14)*. Association for Computing Machinery, New York, NY, USA, 17–25. doi:10.1145/2658537.2658692

[5] Jeremy Gibson Bond. 2014. *Introduction to Game Design, Prototyping, and Development: From Concept to Playable Game with Unity and C#*. Addison-Wesley Professional.

[6] Eva Brandt. 2006. Designing exploratory design games: a framework for participation in Participatory Design. In *Proceedings of the ninth conference on Participatory design: Expanding boundaries in design - Volume 1 (PDC '06)*. Association for Computing Machinery, New York, NY, USA, 57–66. doi:10.1145/1147261.1147271

[7] Eva Brandt, Thomas Binder, and Elizabeth B.-N. Sanders. 2012. Tools and techniques: Ways to engage telling, making and enacting. In *Routledge International Handbook of Participatory Design*. Routledge.

[8] Tone Bratteteig and Ina Wagner. 2016. What is a participatory design result?. In *Proceedings of the 14th Participatory Design Conference: Full papers - Volume 1 (PDC '16)*. Association for Computing Machinery, New York, NY, USA, 141–150. doi:10.1145/2940299.2940316

[9] Jaana Brinck, Teemu Leinonen, Lasse Lipponen, and Mira Kallio-Tavin. 2022. Zones of participation – a framework to analyse design roles in early childhood education and care (ECEC). *CoDesign* 18, 2 (April 2022), 208–226. doi:10.1080/15710882.2020.1812667

[10] Licia Calvi, Jessika Weber-Sabil, Daniel Asmar, and Xavi Socías Perez. 2022. A Framework for Stakeholders' Involvement in Digital Productions for Cultural Heritage Tourism. *Multimodal Technologies and Interaction* 6, 7 (July 2022), 52. doi:10.3390/mti6070052

[11] Colin Camerer, George Loewenstein, and Martin Weber. 1989. The Curse of Knowledge in Economic Settings: An Experimental Analysis. *Journal of Political Economy* 97, 5 (Oct. 1989), 1232–1254. doi:10.1086/261651

[12] Kuo Ping Chang, Yu Wei Huang, Shu Yin Hsueh, Yuh Tyng Chen, Shun Nung Huang, Chien-Hsu Chen, and Sheng-Fen Chien. 2014. HIDDEN LION: a location based app game of sword lion searching. In *Proceedings of the First ACM SIGCHI Annual Symposium on Computer-Human Interaction in Play* (Toronto, Ontario, Canada) *(CHI PLAY '14)*. Association for Computing Machinery, New York, NY, USA, 323–326. doi:10.1145/2658537.2662975

[13] Hongxi Chen, Jia Zhang, Zixuan Guo, Nilufar Baghaei, and Hai-Ning Liang. 2026. XR Intergenerational Exergaming: Exploring the Complementary Roles of VR, MR, and Environmental Context. *IEEE Transactions on Visualization and Computer Graphics* 32, 5 (2026), 4123–4132. doi:10.1109/TVCG.2026.3680711

[14] Si Chen, Xinyue Chen, Zhicong Lu, and Yun Huang. 2023. "My Culture, My People, My Hometown": Chinese Ethnic Minorities Seeking Cultural Sustainability by Video Blogging. *Proc. ACM Hum.-Comput. Interact.* 7, CSCW1, Article 76 (April 2023), 30 pages. doi:10.1145/3579509

[15] Hyungjun Cho, Jiyeon Amy Seo, Jiwon Lee, Chang-Min Kim, and Tek-Jin Nam. 2025. ShamAIn: Designing Superior Conversational AI Inspired by Shamanism. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, 1–18. doi:10.1145/3706598.3714297

[16] Dimitrios Christopoulos, Pavlos Mavridis, Anthousis Andreadis, and John N. Karigiannis. 2011. Using Virtual Environments to Tell the Story: "The Battle of Thermopylae". In *2011 Third International Conference on Games and Virtual Worlds for Serious Applications*. 84–91. doi:10.1109/VS-GAMES.2011.18

[17] Luigina Ciolfi, Daniela Petrelli, Fiona McDermott, Gabriela Avram, and Dick van Dijk. 2015. Co-design to Empower Cultural Heritage Professionals as Technology Designers. In *Empowering Users through Design: Interdisciplinary Studies and Combined Approaches for*

Technological Products and Services, David Bihanic (Ed.). Springer International Publishing, Cham, 213–224. doi:10.1007/978-3-319-13018-7_12

[18] Lucio Tommaso De Paolis, Sofia Chiarello, Carola Gatto, Silvia Liaci, and Valerio De Luca. 2022. Virtual reality for the enhancement of cultural tangible and intangible heritage: The case study of the Castle of Corsano. *Digital Applications in Archaeology and Cultural Heritage* 27 (Dec. 2022), e00238. doi:10.1016/j.daach.2022.e00238

[19] Simone de Sousa Borges, Vinicius H. S. Durelli, Helena Macedo Reis, and Seiji Isotani. 2014. A systematic mapping on gamification applied to education. In *Proceedings of the 29th Annual ACM Symposium on Applied Computing* (Gyeongju, Republic of Korea) *(SAC '14)*. Association for Computing Machinery, New York, NY, USA, 216–222. doi:10.1145/2554850.2554956

[20] Sebastian Deterding, Dan Dixon, Rilla Khaled, and Lennart Nacke. 2011. From game design elements to gamefulness: defining "gamification". In *Proceedings of the 15th International Academic MindTrek Conference: Envisioning Future Media Environments* (Tampere, Finland) *(MindTrek '11)*. Association for Computing Machinery, New York, NY, USA, 9–15. doi:10.1145/2181037.2181040

[21] Francesca Dolcetti. 2025. Crafting Digital Experiences: Embedding Human-Centred and Participatory Design into Archaeological Practice. *J. Comput. Cult. Herit.* 18, 1 (2025), 3:1–3:20. doi:10.1145/3700880

[22] Tao Fan, Hao Wang, and Tobias Hodel. 2023. CICHMKG: A Large-Scale and Comprehensive Chinese Intangible Cultural Heritage Multimodal Knowledge Graph. *Heritage Science* 11, 1 (2023), 115. doi:10.1186/s40494-023-00927-2

[23] Peter Gallert, Colin Stanley, and Kasper Rodil. 2018. Perspectives on Safeguarding Indigenous Knowledge and Intangible Cultural Heritage. In *Proceedings of the Second African Conference for Human Computer Interaction: Thriving Communities* (New York, NY, USA, 2018-12-03) *(AfriCHI '18)*. Association for Computing Machinery, 1–4.

[24] Fengsen Gao, Ke Fang, and Wai Kin (Victor) Chan. 2024. Humanizing Artifacts: An Educational Game For Cultural Heritage Artifacts and History Using Generative AI. In *Companion Proceedings of the 2024 Annual Symposium on Computer-Human Interaction in Play* (Tampere, Finland) *(CHI PLAY Companion '24)*. Association for Computing Machinery, New York, NY, USA, 91–96. doi:10.1145/3665463.3678792

[25] Qing Gong, Ning Zou, Wenjing Yang, Qi Zheng, and Pengrui Chen. 2024. User experience model and design strategies for virtual reality-based cultural heritage exhibition. *Virtual Reality* 28, 2 (March 2024), 69. doi:10.1007/s10055-024-00942-z

[26] Jingyan Guo and Byunghak Ahn. 2023. Tacit Knowledge Sharing for Enhancing the Sustainability of Intangible Cultural Heritage (ICH) Crafts: A Perspective from Artisans and Academics under Craft–Design Collaboration. *Sustainability* 15, 20 (Jan. 2023), 14955. doi:10.3390/su152014955

[27] Yumeng Hou, Sarah Kenderdine, Davide Picca, Mattia Egloff, and Alessandro Adamou. 2022. Digitizing Intangible Cultural Heritage Embodied: State of the Art. *J. Comput. Cult. Herit.* 15, 3, Article 55 (Sept. 2022), 20 pages. doi:10.1145/3494837

[28] Robin Hunicke, Marc LeBlanc, and Robert Zubek. 2004. MDA: A formal approach to game design and game research. In *Proceedings of the AAAI Workshop on Challenges in Game AI*, Vol. 4. San Jose, CA, 1722. https://cdn.aaai.org/Workshops/2004/WS-04-04/WS04-04-001.pdf

[29] Chiara Innocente, Francesca Nonis, Antonio Lo Faro, Rossella Ruggieri, Luca Ulrich, and Enrico Vezzetti. 2024. A Metaverse Platform for Preserving and Promoting Intangible Cultural Heritage. *Applied Sciences* 14, 8 (Jan. 2024), 3426. doi:10.3390/app14083426

[30] Jugkrich Jairassamee and Pongpipat Saithong. 2025. Co-Designing Immersive Experiences for Cultural Heritage: A User-Centric VR Approach for Intelligent Cultural Tourism. *Architecture Image Studies* 6, 3 (Nov. 2025), 1332–1346. doi:10.62754/ais.v6i3.454

[31] Brigitte Jordan and Austin Henderson. 1995. Interaction Analysis: Foundations and Practice. *The Journal of the Learning Sciences* 4 (Jan. 1995), 39–103. doi:10.1207/s15327809jls0401_2

[32] Dominic Kao, Alejandra J. Magana, and Christos Mousas. 2021. Evaluating Tutorial-Based Instructions for Controllers in Virtual Reality Games. *Proc. ACM Hum.-Comput. Interact.* 5, CHI PLAY (2021), 234:1–234:28. doi:10.1145/3474661

[33] Robert S. Kennedy, Norman E. Lane, Kevin S. Berbaum, and Michael G. Lilienthal. 1993. Simulator Sickness Questionnaire: An Enhanced Method for Quantifying Simulator Sickness. *The International Journal of Aviation Psychology* 3, 3 (July 1993), 203–220. doi:10.1207/s15327108ijap0303_3 _eprint: https://doi.org/10.1207/s15327108ijap0303_3.

[34] Eunhye Grace Ko, Rotem Landesman, Jason C Young, Ahmer Arif, Katie Davis, and Angela D. R. Smith. 2025. Domain Experts, Design Novices: How Community Practitioners Enact Participatory Design Values. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, 1–16. doi:10.1145/3706598.3714060

[35] Cuiting Kong. 2024. Digital Diabolo: A Virtual Reality Game for the Presentation of Intangible Cultural Heritage Through Participatory Design. In *Proceedings of the Participatory Design Conference 2024: Situated Actions, Doctoral Colloquium, PDC Places, Communities - Volume 3* (Sibu, Malaysia) *(PDC '24)*. Association for Computing Machinery, New York, NY, USA, 19–23. doi:10.1145/3661456.3666052

[36] Federico Lenzerini. 2011. Intangible Cultural Heritage: The Living Culture of Peoples. *European Journal of International Law* 22, 1 (02 2011), 101–120. arXiv:https://academic.oup.com/ejil/article-pdf/22/1/101/1199591/chr006.pdf doi:10.1093/ejil/chr006

[37] Jiachen Liang, Yue Li, Fanpei Liu, Yu Cheng, Richard Koeck, and Hai-Ning Liang. 2026. Mu-nopoly: Design and Evaluation of a Board Game for Museum Edutainment. *IEEE Transactions on Computational Social Systems* (2026), 1–12. doi:10.1109/TCSS.2025.3641787

[38] Amna Liaqat, Benett Axtell, and Cosmin Munteanu. 2021. Participatory Design for Intergenerational Culture Exchange in Immigrant Families: How Collaborative Narration and Creation Fosters Democratic Engagement. *Proc. ACM Hum.-Comput. Interact.* 5, CSCW1, Article 98 (April 2021), 40 pages. doi:10.1145/3449172

[39] Amna Liaqat, Benett Axtell, and Cosmin Munteanu. 2022. "With a hint she will remember": Collaborative Storytelling and Culture Sharing between Immigrant Grandparents and Grandchildren Via Magic Thing Designs. *Proc. ACM Hum.-Comput. Interact.* 6, CSCW2 (2022), 268:1–268:37. doi:10.1145/3555158

[40] Yilin Liu, Yiming Lin, Rongkai Shi, Yiming Luo, and Hai-Ning Liang. 2021. RelicVR: A Virtual Reality Game for Active Exploration of Archaeological Relics *(CHI PLAY '21)*. Association for Computing Machinery, New York, NY, USA, 326–332. doi:10.1145/3450337.3483507

[41] Zixiao Liu, Shuo Yan, Yu Lu, and Yuetong Zhao. 2022. Generating Embodied Storytelling and Interactive Experience of China Intangible Cultural Heritage "Hua'er" in Virtual Reality. In *Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems* (New York, NY, USA, 2022-04-28) *(CHI EA '22)*. Association for Computing Machinery, 1–7.

[42] Zhicong Lu, Michelle Annett, Mingming Fan, and Daniel Wigdor. 2019. "I feel it is my responsibility to stream": Streaming and Engaging with Intangible Cultural Heritage through Livestreaming. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland Uk) *(CHI '19)*. Association for Computing Machinery, New York, NY, USA, 1–14. doi:10.1145/3290605.3300459

[43] Yuhan Ma, Weiran Zhao, Xiaolin Zhang, and Ze Gao. 2023. Embodied Cognition Guides Virtual-Real Interaction Design to Help Yicheng Flower Drum Intangible Cultural Heritage Dissemination. In *2023 Asia Conference on Cognitive Engineering and Intelligent Interaction (CEII)*. 1–7. doi:10.1109/CEII60565.2023.00009

[44] Marco Mason and Giasemi Vavoula. 2021. Digital Cultural Heritage Design Practice: A Conceptual Framework. *The Design Journal* 24, 3 (May 2021), 405–424. doi:10.1080/14606925.2021.1889738

[45] Francesco Saverio Massari, Pasquale Del Vecchio, and Eva Degl'innocenti. 2022. Past for Future – museums as a digitalized "interaction platform" for value co-creation in tourism destinations. *European Journal of Innovation Management* 27, 5 (Dec. 2022), 1453–1474. doi:10.1108/EJIM-09-2022-0521

[46] Laura Maye and Caroline Claisse. 2023. Co-Design within and between Communities in Cultural Heritage: Current and Open Questions. *Multimodal Technologies and Interaction* 7, 1 (Jan. 2023), 1. doi:10.3390/mti7010001

[47] Michela Mortara, Chiara Eva Catalano, Francesco Bellotti, Giusy Fiucci, Minica Houry-Panchetti, and Panagiotis Petridis. 2014. Learning cultural heritage by serious games. *Journal of Cultural Heritage* 15, 3 (May 2014), 318–325. doi:10.1016/j.culher.2013.04.004

[48] Martha Mosha and Øyvind Eide. 2025. Implementation of Co-design in African Digital Cultural Heritage Public Engagement Tools Development. In *Proceedings of the 5th Biennial African Human Computer Interaction Conference (AfriCHI '25)*. Association for Computing Machinery, New York, NY, USA, 135–140. doi:10.1145/3757232.3757242

[49] Michael Muller and Allison Druin. 2002. Participatory Design: The Third Space in HCI. *Handbook of HCI* (Jan. 2002).

[50] Sanni Mustonen, Siiri Paananen, Emma Eckhoff, and Ashley Colley. 2024. Balancing Playability and Historical Accuracy in Game Design - Developing the Struve Geodetic Arc Mobile Game. *Proc. ACM Hum.-Comput. Interact.* 8, CHI PLAY (2024), 295:1–295:16. doi:10.1145/3677060

[51] Vasiliki Nikolakopoulou and Panayiotis Koutsabasis. 2025. A Framework for Co-designing Mixed Reality Experiences to Enrich Intangible Cultural Heritage in Museums. In *Proceedings of the 3rd International Conference of the ACM Greek SIGCHI Chapter (CHIGreece '25)*. Association for Computing Machinery, New York, NY, USA, 117–125. doi:10.1145/3749012.3749053

[52] Valentina Nisi, Paulo Bala, Vanessa Cesário, Stuart James, Alessio Del Bue, and Nuno Jardim Nunes. 2023. "Connected to the people": Social Inclusion & Cohesion in Action through a Cultural Heritage Digital Tool. *Proc. ACM Hum.-Comput. Interact.* 7, CSCW2, Article 319 (Oct. 2023), 37 pages. doi:10.1145/3610168

[53] Obinna Ogbonnia Otuu and Deepak Sahoo. 2025. How Should We Design Technology With Diverse Stakeholders Who Wish Not to Attend Design Activities Together?. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, 1–14. doi:10.1145/3706598.3714168

[54] Daniela Petrelli and Andrew J. Roberts. 2023. Exploring Digital Means to Engage Visitors with Roman Culture: Virtual Reality vs. Tangible Interaction. *J. Comput. Cult. Herit.* 16, 4 (2023), 73:1–73:18. doi:10.1145/3625367

[55] Sophia Ppali, Ethan Cheung, Alexandra Covaci, Wan-Jou She, and Chee Siang Ang. 2025. Creating with Care: Co-Designing Immersive Experiences through Art-Making with People Living with Dementia. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, 1–18. doi:10.1145/3706598.3714101

[56] Chad Richards, Craig W. Thompson, and Nicholas Graham. 2014. Beyond designing for motivation: the importance of context in gamification. In *Proceedings of the First ACM SIGCHI Annual Symposium on Computer-Human Interaction in Play* (Toronto, Ontario, Canada) *(CHI PLAY '14)*. Association for Computing Machinery, New York, NY, USA, 217–226. doi:10.1145/2658537.2658683

[57] Richard M. Ryan. 1982. Control and information in the intrapersonal sphere: An extension of cognitive evaluation theory. *Journal of Personality and Social Psychology* 43, 3 (1982), 450–461. doi:10.1037/0022-3514.43.3.450

[58] Elizabeth B.-N. Sanders and Pieter Jan Stappers. 2008. Co-creation and the new landscapes of design. *CoDesign* 4, 1 (2008), 5–18. arXiv:https://doi.org/10.1080/15710880701875068 doi:10.1080/15710880701875068

[59] Brenda Saris, Stephanie Doyle, and Judith Loveridge. 2023. Analysing Creative Design Process: A Set of Tools to Understand Activity in its Socio-cultural and Historic Context. *International Journal of Art & Design Education* 42, 3 (2023), 439–453. doi:10.1111/jade.12467

[60] Thomas Schubert, Frank Friedmann, and Holger Regenbrecht. 2001. The Experience of Presence: Factor Analytic Insights. *Presence: Teleoperators and Virtual Environments* 10, 3 (06 2001), 266–281. arXiv:https://direct.mit.edu/pvar/article-pdf/10/3/266/1623697/105474601300343603.pdf doi:10.1162/105474601300343603

[61] Elmedin Selmanović, Selma Rizvic, Carlo Harvey, Dusanka Boskovic, Vedad Hulusic, Malek Chahin, and Sanda Sljivo. 2020. Improving Accessibility to Intangible Cultural Heritage Preservation Using Virtual Reality. *J. Comput. Cult. Herit.* 13, 2, Article 13 (May 2020), 19 pages. doi:10.1145/3377143

[62] Hyeseung Shim, Eun Seo Kim, Kyung Taek Oh, Chung-Kon Shi, and Jaehong Ahn. 2024. Diving into a Heritage Metaverse: Learning from End User-driven Experiences and Perspectives to Enhance Virtual Heritage Interpretation. *Proc. ACM Hum.-Comput. Interact.* 8, CSCW1 (2024), 45:1–45:28. doi:10.1145/3637322

[63] Maria Skublewska-Paszkowska, Marek Milosz, Pawel Powroznik, and Edyta Lukasik. 2022. 3D technologies for intangible cultural heritage preservation—literature review for selected databases. *Heritage Science* 10, 1 (Jan. 2022), 3. doi:10.1186/s40494-021-00633-x Publisher: Nature Publishing Group.

[64] Kihoon Son, DaEun Choi, Tae Soo Kim, and Juho Kim. 2024. Demystifying Tacit Knowledge in Graphic Design: Characteristics, Instances, Approaches, and Guidelines. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*. Association for Computing Machinery, New York, NY, USA, 1–18. doi:10.1145/3613904.3642886

[65] Marc Steen. 2013. Co-Design as a Process of Joint Inquiry and Imagination. *Design Issues* 29, 2 (2013), 16–28. doi:10.1162/DESI_a_00207

[66] Chen Tao and Xinghua Hong. 2025. A three-dimensional visualization model for intangible heritage of tie-dye handicrafts. *npj Heritage Science* 13, 1 (Dec. 2025), 648. doi:10.1038/s40494-025-02223-7 Publisher: Nature Publishing Group.

[67] Anastasios Theodoropoulos and Angeliki Antoniou. 2022. VR Games in Cultural Heritage: A Systematic Review of the Emerging Fields of Virtual Reality and Culture Games. *Applied Sciences* 12, 17 (Jan. 2022), 8476. doi:10.3390/app12178476

[68] UNESCO. n.d.. What is Intangible Cultural Heritage? https://ich.unesco.org/en/what-is-intangible-heritage-00003

[69] ICH UNESCO. 2003. Intangible Heritage Domains in the 2003 Convention. (2003). https://ich.unesco.org/en/intangible-heritage-domains-00052

[70] Maria Vayanou, Akrivi Katifori, and Yannis Ioannidis. 2021. Perspective Sharing in Culture Group Games: Passing Around the Social Mediator Role. *Proc. ACM Hum.-Comput. Interact.* 5, CHI PLAY (2021), 249:1–249:24. doi:10.1145/3474676

[71] John Vines, Rachel Clarke, Peter Wright, John McCarthy, and Patrick Olivier. 2013. Configuring participation: on how we involve people in design. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '13)*. Association for Computing Machinery, New York, NY, USA, 429–438. doi:10.1145/2470654.2470716

[72] Yingna Wang, Qingqin Liu, Xiaoying Wei, and Mingming Fan. 2025. Facilitating Daily Practice in Intangible Cultural Heritage through Virtual Reality: A Case Study of Traditional Chinese Flower Arrangement. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, 1–17. doi:10.1145/3706598.3713409

[73] Alice Willatt, Stuart Iain Gray, Helen Manchester, Tot Foster, and Kirsten Cater. 2024. An Intersectional Lifecourse Lens and Participatory Methods as the Foundations for Co-Designing with and for Minoritised Older Adults. *Proc. ACM Hum.-Comput. Interact.* 8, CSCW1, Article 18 (April 2024), 29 pages. doi:10.1145/3637295

[74] Haorui Wu and Chaoping Hou. 2019. Utilizing co-design approach to identify various stakeholders' roles in the protection of intangible place-making heritage: The case of Guchengping Village. *Disaster Prevention and Management* 29, 1 (March 2019), 22–35. doi:10.1108/DPM-09-2018-0291

[75] Wenge Xu, Hai-Ning Liang, Kangyou Yu, and Nilufar Baghaei. 2021. Effect of Gameplay Uncertainty, Display Type, and Age on Virtual Reality Exergames. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) *(CHI '21)*. Association for Computing Machinery, New York, NY, USA, Article 439, 14 pages. doi:10.1145/3411764.3445801

[76] Wenge Xu, Hai-Ning Liang, Kangyou Yu, Shaoyue Wen, Nilufar Baghaei, and Huawei Tu. 2023. Acceptance of Virtual Reality Exergames Among Chinese Older Adults. *International Journal of Human–Computer Interaction* 39, 5 (2023), 1134–1148. arXiv:https://doi.org/10.1080/10447318.2022.2098559 doi:10.1080/10447318.2022.2098559

[77] Ruixuan Yang, Yuzhi Li, Yiting Wang, Qiaolian Zhu, Ning Wang, Ying Song, Feng Tian, and Haojun Xu. 2024. Enhancing the Sustainability of Intangible Cultural Heritage Projects: Obtaining Efficient Digital Skills Preservation through Binocular Half Panoramic VR Maps. *Sustainability* 16, 13 (Jan. 2024), 5281. doi:10.3390/su16135281

[78] Naci Yastikli. 2007. Documentation of Cultural Heritage Using Digital Photogrammetry and Laser Scanning. *Journal of Cultural Heritage* 8, 4 (2007), 423–427. doi:10.1016/j.culher.2007.06.003

[79] Jian Yu, Zhan Wang, Yifan Cao, Hao Cui, and Wei Zeng. 2025. Centennial Drama Reimagined: An Immersive Experience of Intangible Cultural Heritage through Contextual Storytelling in Virtual Reality. *J. Comput. Cult. Herit.* 18, 1 (2025), 11:1–11:22. doi:10.1145/3705613

[80] Suki Yuanxin Zhang, R. Achintha M. Abayasiri, Yanzhuo Yang, Hai-Ning Liang, Wenge Xu, and Nilufar Baghaei. 2025. Impact of Visual Embellishments on Player Experience in Virtual Reality Exergames. In *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*. Association for Computing Machinery, New York, NY, USA, Article 334, 6 pages. doi:10.1145/3706599.3721252

[81] Ramadanova Zhanna. 2020. Digitization of Intangible Cultural Heritage as a method to save and actualize it on the example of Kazakh folk dance. In *Proceedings of the 6th International Conference on Engineering & MIS 2020* (Almaty, Kazakhstan) *(ICEMIS'20)*. Association for Computing Machinery, New York, NY, USA, Article 16, 7 pages. doi:10.1145/3410352.3410748

[82] Ke Zhao, Ruiqi Chen, Xiaziyu Zhang, Chenxi Wang, Siling Chen, Xiaoguang Wang, Yujue Wang, and Xin Tong. 2025. Immersive Biography: Supporting Intercultural Empathy and Understanding for Displaced Cultural Objects in Virtual Reality. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. ACM, Yokohama Japan, 1–17. doi:10.1145/3706598.3714303

[83] Yiqing Zhao, Yaning Li, Tianchen Dai, Carla Sedini, Xue Wu, Weile Jiang, Ji Li, Kaiyi Zhu, Binqing Zhai, Meng Li, and Ray Lc. 2025. Virtual reality in heritage education for enhanced learning experience: a mini-review and design considerations. *Frontiers in Virtual Reality* 6 (March 2025). doi:10.3389/frvir.2025.1560594

[84] Yuting Zhu and Xin He. 2025. Factors influencing user acceptance of virtual reality in preserving Wuhu iron painting from the unified theory of acceptance and use of technology perspective. *Scientific Reports* 15, 1 (July 2025), 22714. doi:10.1038/s41598-025-08310-w

A Pre-Workshop Questionnaire

A.1 Demographic Information

Single choice: select one option for each item.

- Age
- Gender
- Education level
- Occupation
- Length of local residence

A.2 Cultural Experience Dimension (5-point Likert)

Unless otherwise noted, use a 5-point scale appropriate to the item.

CE1. I have personally participated in the "Pahetong" activity.
CE2. I have heard of the "Pahetong" cultural practice before.
CE3. I understand how the "Pahetong" activity is performed.
CE4. I am familiar with the historical or cultural background of "Pahetong".
CE5. I have practical experience with "Pahetong" or similar agricultural practices in real-life settings.

A.3 Creative Design Dimension (5-point Likert)

Creative Self-Efficacy Scale. Please indicate agreement on a 5-point scale.

CSE1. I am confident in my ability to solve problems creatively.
CSE2. I consider myself good at coming up with new ideas.
CSE3. I am skilled at developing my own ideas based on others' input.
CSE4. I am good at finding new ways to solve problems.

Additional items:

CD1. To what extent do you have experience in game design?
CD2. How familiar are you with paper prototyping in game design?

A.4 VR Experience Dimension (5-point Likert)

VR1. How familiar are you with virtual reality (VR) interactions?
VR2. How frequently have you used VR devices?
VR3. How often have you played VR games or experienced VR-based cultural content?
VR4. How interested are you in VR-based cultural games?

B Workshop Participant Expertise Profiles

Table 4. Participant-level self-rated expertise profiles. Scores are raw mean values from 1–5 Likert-scale screening items. Participants are grouped by workshop configuration.

Participant	Culture	Design	VR	Relevant background
Culture-oriented workshop				
C1	3.60	5.00	3.75	Local resident
C2	2.60	3.00	2.75	Local resident for more than 10 years
C3	4.40	3.67	2.00	Local resident with prior *Pahetong* experience

Table 4. Participant-level self-rated expertise profiles (continued).

Participant	Culture	Design	VR	Relevant background
C4	3.80	5.00	2.00	Local resident with prior *Pahetong* experience
C5	1.80	3.33	2.25	Local resident for more than 10 years
C6	2.80	3.17	1.75	Local resident with prior *Pahetong* experience
Design-oriented workshop				
D1	1.20	4.17	2.75	Theatre producer
D2	2.00	3.83	3.00	Designer
D3	1.00	4.50	4.25	PhD student in design
D4	1.60	4.00	3.00	Fashion entrepreneur
D5	1.20	4.83	4.75	Student
VR-oriented workshop				
V1	1.00	4.17	4.75	VR producer
V2	1.00	3.17	4.00	Student
V3	1.00	3.17	4.50	PhD student in VR health
V4	1.40	3.33	3.00	Student

C Co-design Workshop Protocol

Table 5. Replicable workshop protocol used across the three stakeholder-configured co-design workshops.

Phase	Time	Materials	Procedure	Output
Check-in	5 min	Consent forms; name labels; sticky notes; pens; voting dots; prototyping materials	Confirm consent; distribute materials; introduce workshop structure.	Participants ready for workshop.
Context and shared references	15 min	*Pahetong* slides/images; VR primer; Beat Saber and Moss examples	Introduce *Pahetong*; explain VR interaction; present reference VR titles.	Shared cultural and VR references.
Game-design decomposition	15 min	MDA slides; Experience–Play–Rules prompt sheet; reference game examples	Explain simplified MDA terms; guide participants to identify experience, play, and rules in examples.	Shared design vocabulary.

Continued on next page

Table 5. Replicable workshop protocol used across the three stakeholder-configured co-design workshops (continued).

Phase	Time	Materials	Procedure	Output
Divergent ideation and guided convergence	30 min	Sticky notes; word cards; pens; voting dots; E→P→R prompts	Generate *Pahetong*-related keywords; run two-word sentence-making; dot-vote core concepts; guide convergence through Experience, Play, and Rules.	Selected core concept; preliminary role, goal, interactables, input mappings, feedback, and end condition.
Paper prototyping and enactment	20 min	Large paper sheets; pens; tokens; desktop props; hand-drawn scene elements	Build a minimal paper prototype; clarify resources, actions, rules, and feedback; briefly enact the core loop.	Runnable paper prototype and core rule structure.
Reflection and consolidation	5 min	Prototype materials; facilitator notes; audio/video recording	Summarize final concept; invite comments on feasibility, cultural fit, and missing elements.	Consolidated design notes and participant explanations.

D Self-constructed questionnaire items

D.1 Cultural Understanding items

Participants rated the following items on a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree).

(1) This game helped me imagine what it would feel like to take part in the "Pahetong" activity.
(2) This game helped me understand the concrete steps of how "Pahetong" is performed.
(3) This game helped me understand the historical background or cultural meaning of "Pahetong".
(4) This game made me more interested in learning more about "Pahetong" beyond the game itself.

D.2 Perceived Completeness items

Participants rated the following items on a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree).

(1) I think this game provided a complete and coherent gameplay experience.
(2) I think the visuals and audio of this game feel polished and high in production quality.
(3) The interaction in this game felt smooth to me.
(4) I could clearly understand the meaning of the feedback effects in this game.
(5) This game felt rather rough or unfinished to me (reverse-coded).

E Semi-structured interview guide

After playing the three VR game versions and completing the post-questionnaire, participants carried out a ranking task. During the ranking process, the interviewer invited participants to briefly explain the reasons for their rankings while they were ordering and adjusting the three versions. They were asked to rank the three versions on four criteria:

(1) which version best helped them imagine what taking part in Pahetong is like,
(2) which version was most enjoyable to play,

(3) which version felt clearest and easiest to understand, and
(4) which version they would most recommend overall.

After the ranking task, a short semi-structured interview was conducted with the following guiding questions:

(1) Based on the games you just played, how would you describe your current understanding of the "Pahetong" activity?
(2) For example, when and where do you think it usually takes place, and in what form is it carried out?
(3) Which game version, and which specific elements in the games, contributed most to this understanding?
(4) *Interviewer briefly introduces the real-world Pahetong activity.*
After hearing my introduction, in what ways does it match or differ from what you experienced in the games?
(5) Do you have any suggestions for improving the Pahetong games? Which parts do you think could be improved, and how would you optimise them?

F Workshop Validation Questionnaire Items.

After playing the prototype corresponding to their workshop, participants rated the following statements on a 7-point Likert scale (1 = *Strongly disagree*, 7 = *Strongly agree*):

(1) The prototype overall matches the game concept we designed in the workshop.
(2) The core gameplay mechanics in the prototype are consistent with what we discussed in the workshop.
(3) The way *Pahetong* is represented in the prototype largely reflects what we intended to convey in the workshop.
(4) The prototype is missing some elements that we considered important in the workshop (reverse-coded).

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